



Enhancing information discriminant analysis: Feature extraction with linear statistical model and information-theoretic criteria[☆]

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ABSTRACT

In this paper, we develop a novel feature transformation method for supervised linear dimensionality reduction. Existing methods, e.g., Information Discriminant Analysis (IDA), estimate the first and second order statistics of the data in the original high-dimensional space, and then design the transformation matrix based on the information-theoretic criteria. Unfortunately, such transformation methods are sensitive to the accuracy of the statistics estimation. To overcome this disadvantage, our method describes the statistical structure of the transformed low-dimensional subspace via a linear statistical model, which can reduce the number of unknown parameters, while simultaneously maximizes the mutual information (MI) between the transformed data and their class labels, which can ensure the between-class separability according to the information theory. The key idea is that we seek the optimal model parameters, including the transformation matrix, via the joint optimization of MI function and log-likelihood function, therefore, this method can not only reduce the estimation errors but also maximize the between-class separability. Experimental results based on synthetic dataset and benchmark datasets demonstrate the better performance of our method over other related methods.

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1. Introduction

With the rapid development of science and technology and the emergence of large-scale high-dimensional data, people urgently need to classify high-dimensional data accurately and efficiently. In high-dimensional data, some features are redundant for classification, which not only increase the cost of data processing, but also seriously affect the accuracy of classification. Therefore, dimensionality reduction becomes an important preprocessing step for high-dimensional data classification. There are two major categories of dimensionality reduction methods: feature selection and feature transform [1]. Feature selection methods only keep useful features and discard others. Typical examples include Fisher [2], Relief [3], exhaustive method [2] and so on. Feature transform methods construct new features from the original variables. Typical examples include Principal Component Analysis (PCA) [4], Linear Discriminant Analysis (LDA) [5,6], Median-Mean Line based Discriminant Analysis (MMLDA) [34], Information Discriminant

Analysis (IDA) [7], Independent Component Analysis (ICA) [25–30].

In this paper we focus on the feature transformation design for high-dimensional data classification. A feature transform method is usually coupled with an appropriate criterion, which measures the joint “importance” of a set of features. If the criterion is differentiable with respect to the parameters of the transform, and the transform is smooth, it is possible to learn the transform by optimizing the criterion. A well-known feature transform method is LDA, which maximizes the between-class scattering while minimizing the within-class scattering of the transformed data [5]. An advantage of LDA is that the transformation matrix can be found analytically, thereby avoiding numerical optimization. However, the dimensionality of the transformed space in LDA can only be less than the number of data classes, which greatly restricts its performance and applications. To overcome this disadvantage and improve the robustness on outliers, MMLDA is proposed based on the traditional LDA method in [34]. The median mean line is introduced in MMLDA to substitute the original mean vector in LDA, which is the line through the median sample and mean vector. Due to the median sample is nearly stable for the outliers, the median mean line is more robust than the mean vector to outliers. In addition, since the new defined between-class scattering matrix in MMLDA is non-singular for any transformed subspace size, then there is no restriction for the transformed space size any more in the MMLDA method. Besides the

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improved versions of LDA, some feature transformation methods based on the information-theoretic criteria can also improve the performance of the traditional LDA [8,14]. Tao et al. [8] designs the transformation matrix by maximizing the mutual information (MI) between the transformed data and their class labels [1,7,9–11,24]. According to the theoretic analysis [12,13], the larger the MI is, the tighter the bound of Bayes error rate will be. IDA [7] and the work of [14] are two typically supervised dimensionality reduction methods based on the information-theoretic criteria. Since the MI is very difficult to calculate directly for non-Gaussian distributions, IDA approximates the entropy of the Gaussian mixture model (GMM) with the entropy of a global Gaussian distribution, and provides an approximate expression to the Shannon MI. In the work of [14], an analytic and explicit expression for the gradient of the Shannon MI with respect to the transformation matrix is given for any distribution, thus the Shannon MI can be maximized directly without any approximation. For convenience, the method in [14] is referred to as mutual information maximization (MIM) in this paper. Both methods estimate the first and second order statistics of the data in the high-dimensional space, and then design the transformation matrix based on the information-theoretic criteria. However, in practice, especially in the case of limited data, it is difficult to accurately estimate these statistics in the original high-dimensional space. Thus the transformation matrix obtained may be nonoptimal, which will affect the final classification performance.

This paper proposes a novel supervised dimensionality reduction method based on linear statistical model and information-theoretic criterion, conveniently called enhancing information discriminant analysis (EIDA). Our method describes the statistical structure of the transformed low-dimensional subspace via a linear statistical model, and utilizes the MI based information-theoretic criterion to further maximize the between-class separability. Inspired by IDA, our method seeks the optimal model parameters, including transformation matrix, via the joint optimization of MI function and log-likelihood function. The main contributions of this paper are summarized in the following.

1) In our linear statistical model, the original high-dimensional sample is represented as the linear combination of low-dimensional feature and additive measurement noise. based on such a linear model, the statistics can be estimated in the low-dimensional subspace. therefore, the estimation errors can be reduced due to the lower freedom of unknown parameters, compared with those got from the original high-dimensional data space, especially in the case of limited data. 2) in our optimization algorithm, the joint optimization designs the transformation matrix and estimates the data statistics simultaneously. Therefore, we can not only ensure the between-class separability, but also describe the observed data as accurately as possible.

Here we give Table 1 to list all notations in our EIDA model to help readers to understand the dimensions of matrices and definitions of notations clearly.

The remainder of the paper is organized as follows: Section 2 is a brief overview of Bayesian classification, MI, and IDA. Section 3 introduces EIDA. A complete algorithm is developed and presented in this section with some detailed derivations given in Appendix. Section 4 describes some related methods. In Section 5, we validate the performance of our method on synthetic dataset and some benchmark datasets. Section 6 summarizes the paper by presenting some concluding remarks.

Table 1

List of notations in the EIDA model.

d : Dimensionality of original data
m : Dimensionality of low-dimensional latent feature
N_k : Data size of training sample for class k
n : Number of total samples (include training samples and test samples)
$\mathbf{x} \in \mathbb{R}^{d \times 1}$: Original data
$\mathbf{y} \in \mathbb{R}^{m \times 1}$: Low-dimensional latent feature of $\mathbf{x}$
$\mathbf{A} \in \mathbb{R}^{d \times m}$: Transformation matrix
$\mathbf{e} \in \mathbb{R}^{d \times 1}$: Measurement noise
$\Psi_k \in \mathbb{R}^{d \times d}$: Diagonal matrix of the noise covariance on class k
$\mathbf{F}_k \in \mathbb{R}^{d \times d}$: Prior precision matrix for original data from class k
$\mathbf{x}_{ki} \in \mathbb{R}^{d \times 1}$: i th data from class k
$\mathbf{y}_{ki} \in \mathbb{R}^{m \times 1}$: Low-dimensional latent feature of $\mathbf{x}_{ki}$
$\mu_{xk} \in \mathbb{R}^{d \times 1}$: Prior mean vector for original data from class k
$\Sigma_{xk} \in \mathbb{R}^{d \times d}$: Prior covariance matrix for original data from class k
$\mu_{yk} \in \mathbb{R}^{m \times 1}$: Condition posterior mean vector for latent feature from class k
$\Sigma_{yk} \in \mathbb{R}^{m \times m}$: Condition posterior covariance matrix for latent feature from class k
$\mu_k \in \mathbb{R}^{d \times 1}, \mu \in \mathbb{R}^{d \times 1}: \mu_k = \mathbf{F}_k \mu_{xk}, \mu = \sum_{k=1}^K w_k \mu_k$
$\Omega_k \in \mathbb{R}^{d \times d}, \Omega \in \mathbb{R}^{d \times d}: \Omega_k = \mathbf{F}_k \Sigma_{xk} \mathbf{F}_k, \Omega = \sum_{k=1}^K w_k [\Omega_k + (\mu_k - \mu)(\mu_k - \mu)^T]$
$\mathbf{O}_k \in \mathbb{R}^{d \times d}, \mathbf{O} \in \mathbb{R}^{d \times d}: \mathbf{O}_k = \mathbf{A}(\mathbf{A}^T \Omega_k \mathbf{A})^{-1} \mathbf{A}^T, \mathbf{O} = \mathbf{A}(\mathbf{A}^T \Omega \mathbf{A})^{-1} \mathbf{A}^T$

2. Background

2.1. Bayesian classification

The Bayesian classifier [2] is often used to evaluate the performance of a supervised dimensionality reduction method. For a discrete-valued class variable C and a continuous feature variable $\mathbf{Y} \in \mathbb{R}^{m \times 1}$, let $p(c = k) \triangleq w_k$ be the prior distribution for $\forall k = 1, 2, \dots, K$ with K denoting the number of classes, and $p(\mathbf{y}|k) = N(\mathbf{y}; \mu_{yk}, \Sigma_{yk})$ be the class-conditional probability density function (PDF), where $N(\mathbf{y}; \mu_{yk}, \Sigma_{yk})$ denotes the Gaussian distribution for variable $\mathbf{y}$ with mean vector μ_{yk} and covariance matrix Σ_{yk} . According to Bayes rule, the posterior probability $p(k|\mathbf{y})$ can be expressed as $p(k|\mathbf{y}) = \frac{p(\mathbf{y}|k)w_k}{p(\mathbf{y})}$, where $p(\mathbf{y}) = \sum_{k=1}^K p(\mathbf{y}|k)w_k$. The Bayesian classifier will assign $\mathbf{y}$ to class t if

$$t = \arg \max_{k: 1, 2, \dots, K} p(k|\mathbf{y}) \quad (1)$$

The above formula can be further expressed as

$$t = \arg \max_{k: 1, 2, \dots, K} \ln[p(\mathbf{y}|k)w_k] \\ = \arg \max_{k: 1, 2, \dots, K} \left[-\frac{m}{2} \ln(2\pi) - \frac{1}{2} \ln |\Sigma_{yk}| - \frac{1}{2} (\mathbf{y} - \mu_{yk})^T \Sigma_{yk}^{-1} (\mathbf{y} - \mu_{yk}) + \ln(w_k) \right] \quad (2)$$

where T denotes the transpose of a vector or a matrix. If the covariance matrices for all of the classes are identical, i.e., $\Sigma_{yk} = \Sigma_y, \forall k = 1, 2, \dots, K$. In this case, from Eq. (2), we have

$$t = \arg \max_{k: 1, 2, \dots, K} \left[\mu_{yk}^T \Sigma_y^{-1} \mathbf{y} - \frac{1}{2} \mu_{yk}^T \Sigma_y^{-1} \mu_{yk} + \ln(w_k) \right] \quad (3)$$

where the resulting discriminant functions are linear. A Bayesian classifier that uses linear discriminant functions is called a linear Bayesian classifier.

Another simple case arises when the covariance matrices are different for each category, then, Eq. (2) can be written as

$$t = \arg \max_{k: 1, 2, \dots, K} \left\{ -\frac{1}{2} \mathbf{y}^T \Sigma_{yk}^{-1} \mathbf{y} + \mu_{yk}^T \Sigma_{yk}^{-1} \mathbf{y} - \frac{1}{2} \mu_{yk}^T \Sigma_{yk}^{-1} \mu_{yk} - \frac{1}{2} \ln |\Sigma_{yk}| + \ln(w_k) \right\} \quad (4)$$

where the resulting discriminant functions are inherently quadratic. We call the classifier a quadratic Bayesian classifier. Consequently, the Bayes error rate can be written as

$$P_e = 1 - \int p(\mathbf{y}) \max_k p(k|\mathbf{y}) d\mathbf{y} \quad (5)$$

2.2. Mutual information

According to Shannon's definition [15], the entropy of the class label is expressed in terms of class prior probabilities $H(C) = -\sum_c p(c) \log(p(c))$. After having observed a feature vector $\mathbf{y}$, the uncertainty of the class identity is now the conditional entropy $H(C|\mathbf{Y}) = -\int_{\mathbf{y}} \sum_c p(c, \mathbf{y}) \log(p(c|\mathbf{y})) d\mathbf{y}$. Then the mutual information between C and $\mathbf{Y}$, i.e., $I(C; \mathbf{Y})$, can be defined as

$$I(C; \mathbf{Y}) = H(C) - H(C|\mathbf{Y}) = -\int_{\mathbf{y}} \sum_c p(c, \mathbf{y}) \log\left(\frac{p(c, \mathbf{y})}{p(c)p(\mathbf{y})}\right) d\mathbf{y} \quad (6)$$

From Eq. (6), when class variable C and feature variable $\mathbf{Y}$ are independent, the joint density $p(c, \mathbf{y})$ can be factored as a product of marginal densities, i.e., $p(c, \mathbf{y}) = p(c)p(\mathbf{y})$. In this case, the MI equals zero. Therefore, MI can measure the dependence between variables. The larger the MI is, the stronger the dependence will be. Based on the mathematical derivation, the Bayes error satisfies the following inequality [12,13]

$$\frac{H(C) - I(\mathbf{Y}; C) - 1}{\log^K} \leq P_e \leq \frac{1}{2} \{H(C) - I(\mathbf{Y}; C)\} \quad (7)$$

As shown in Eq. (7), both the upper bound and the lower bound on the Bayes error are minimized when the MI $I(C; \mathbf{Y})$ is maximized. Therefore, the MI can be used as an evaluation criterion for supervised dimensionality reduction [1,8,16,24].

2.3. Information discriminant analysis

Suppose $\mathbf{x} \in \mathbb{R}^{d \times 1}$ represents the high-dimensional data, in the GMM data model, the data distribution in each class is modeled as a single Gaussian, i.e., $p(\mathbf{x}|k) = N(\mathbf{x}; \boldsymbol{\mu}_{\mathbf{x}k}, \boldsymbol{\Sigma}_{\mathbf{x}k})$, thus the global data density can be written as $p(\mathbf{x}) = \sum_{k=1}^K w_k p(\mathbf{x}|k)$. We seek a transformation matrix $\Phi \in \mathbb{R}^{m \times d} (m < d)$ such that the transformed data $\mathbf{y} = \Phi \mathbf{x}$ is the most informative in identifying the underlying class label c . Then the information-theoretic criterion [7,14,17,18] can be adopted as

$$\max_{\Phi} I(C; \mathbf{Y}) \quad (8)$$

where $I(C; \mathbf{Y})$ denotes the Shannon MI. According to the properties of Gaussian distribution, we can obtain that $p(\mathbf{y}|k) = p(\mathbf{y}; \Phi \boldsymbol{\mu}_{\mathbf{x}k}, \Phi \boldsymbol{\Sigma}_{\mathbf{x}k} \Phi^T)$, and $p(\mathbf{y}) = \sum_{k=1}^K w_k p(\mathbf{y}|k)$. Thus the Shannon MI can be expressed as

$$\begin{aligned} I(C; \mathbf{Y}) &= H(C) - H(C|\mathbf{Y}) = H(\mathbf{Y}) - H(\mathbf{Y}|C) \\ &= H(\mathbf{Y}) - \frac{1}{2} \sum_{k=1}^K w_k \log[(2\pi e)^m \det(\Phi \boldsymbol{\Sigma}_{\mathbf{x}k} \Phi^T)] \end{aligned} \quad (9)$$

where e denotes the base of natural logarithms. Since $p(\mathbf{y})$ is a GMM whose entropy cannot be expressed analytically, IDA [7] approximates $H(\mathbf{Y})$ with a global Gaussian entropy with the same covariance matrix as $p(\mathbf{y})$. The approximated MI can be expressed as

$$I_{DA}(C; \mathbf{Y}) = \frac{1}{2} \left[\log |\Phi \boldsymbol{\Sigma}_{\mathbf{x}} \Phi^T| - \sum_{k=1}^K w_k \log |\Phi \boldsymbol{\Sigma}_{\mathbf{x}k} \Phi^T| \right] \quad (10)$$

where $\boldsymbol{\Sigma}_{\mathbf{x}} = \sum_{k=1}^K w_k [\boldsymbol{\Sigma}_{\mathbf{x}k} + (\boldsymbol{\mu}_{\mathbf{x}k} - \boldsymbol{\mu}_{\mathbf{x}})(\boldsymbol{\mu}_{\mathbf{x}k} - \boldsymbol{\mu}_{\mathbf{x}})^T]$ is the prior covariance matrix for $\mathbf{x}$ and $\boldsymbol{\mu}_{\mathbf{x}} = \sum_{k=1}^K w_k \boldsymbol{\mu}_{\mathbf{x}k}$ is the prior mean. The gradient of $I_{DA}(C; \mathbf{Y})$ with respect to the transformation matrix Φ can be written as

$$\frac{\partial I_{DA}(C; \mathbf{Y})}{\partial \Phi} = (\Phi \boldsymbol{\Sigma}_{\mathbf{x}} \Phi^T)^{-1} \Phi \boldsymbol{\Sigma}_{\mathbf{x}} - \sum_{k=1}^K w_k (\Phi \boldsymbol{\Sigma}_{\mathbf{x}k} \Phi^T)^{-1} \Phi \boldsymbol{\Sigma}_{\mathbf{x}k} \quad (11)$$

Then the optimization of $I_{DA}(C; \mathbf{Y})$ can be solved directly via the gradient descent method.

3. Enhancing information discriminant analysis

The performance of IDA relies on the accurate estimation of the means and covariance matrices of the high-dimensional data. However, in practice, the data available is limited, while the degree of freedom of the covariance matrix is usually large, hence estimating these quantities directly in the high-dimensional space is usually inaccurate. To overcome this problem, this paper presents a new supervised dimensionality reduction method based on linear statistical model and information-theoretic criterion. The basic idea will be introduced in the following.

3.1. Optimization problem

Assuming that the high-dimensional data is generated via the following linear statistical model

$$\mathbf{x} = \mathbf{A}\mathbf{y} + \boldsymbol{\varepsilon} \quad (12)$$

where $\mathbf{x} \in \mathbb{R}^{d \times 1}$ represents the original data from class k , $\mathbf{A} \in \mathbb{R}^{d \times m}$ is the transformation matrix with $m < d$, $\mathbf{y} \in \mathbb{R}^{m \times 1}$ denotes the low-dimensional latent feature of $\mathbf{x}$, and $\boldsymbol{\varepsilon} \in \mathbb{R}^{d \times 1}$ is the measurement noise. Thus the original high-dimensional data $\mathbf{x}$ can be viewed as the linear mapping from a latent feature $\mathbf{y}$ with the added noise $\boldsymbol{\varepsilon}$. Here we assume Gaussian prior on $\mathbf{y}$ and $\boldsymbol{\varepsilon}$, i.e., $p(\mathbf{y}|k) = N(\mathbf{y}; \mathbf{0}, \mathbf{I})$ and $p(\boldsymbol{\varepsilon}|k) = N(\boldsymbol{\varepsilon}; \mathbf{0}, \boldsymbol{\Psi}_k)$, where $\boldsymbol{\Psi}_k$ is a diagonal matrix, representing the noise variance on class k . We introduce such a linear statistical model for each class of the data, in which the transformation matrix $\mathbf{A}$ is shared, but the measurement noises' variance matrixes are different for the different classes. Such statistical model is similar to the factor analysis (FA) model in [19] with the loading matrix (referred to as the transformation matrix in our method) $\mathbf{A}$ sharing among all classes. In the traditional FA model [19], the noise variance matrix $\boldsymbol{\Psi}_k$ is also assumed to be a diagonal matrix. The reason may be to simplify the statistical model and to avoid overfitting for parameter learning. In our method, since the transformation matrix $\mathbf{A}$ is shared, we must use the class specific noise variance matrix $\boldsymbol{\Psi}_k$ to describe the class specific data distribution.

For convenience, we define $\boldsymbol{\Psi} = \{\boldsymbol{\Psi}_k\}_{k=1}^K$. We want to find the model parameters $\{\mathbf{A}, \boldsymbol{\Psi}\}$ which can not only maximize the separability between the latent features from different classes but also describe the statistical structure of observed data accurately. That is the following optimization problem

$$\max_{\mathbf{A}, \boldsymbol{\Psi}} I_{\mathbf{A}, \boldsymbol{\Psi}}(C, \mathbf{Y}) + \lambda L_{\mathbf{A}, \boldsymbol{\Psi}}(\mathbf{X}) \quad (13)$$

where λ denotes the weighting factor, $\mathbf{X}$ denotes the observed dataset, $I_{\mathbf{A}, \boldsymbol{\Psi}}(C, \mathbf{Y})$ represents the MI function between latent feature variable $\mathbf{Y}$ and class variable C , $L_{\mathbf{A}, \boldsymbol{\Psi}}(\mathbf{X})$ represents the log-likelihood function of the linear statistical model. Expressions for log-likelihood function and MI function are described in the following.

According to the definition of linear statistical model and the properties of Gaussian distribution, the prior distribution of the

original data from class k can be expressed as $p(\mathbf{x}|k) = N(\mathbf{x}; \mathbf{0}, \mathbf{F}_k^{-1})$ with $\mathbf{F}_k = (\mathbf{A}\mathbf{A}^T + \boldsymbol{\Psi}_k)^{-1}$ and $k = 1, 2, \dots, K$. Thus the log-likelihood function of the entire dataset can be expressed as

$$L_{\mathbf{A}, \boldsymbol{\Psi}}(\mathbf{X}) = \sum_{k=1}^K \sum_{i=1}^{N_k} \ln P(\mathbf{x}_{ki}) \\ = -\frac{1}{2} \sum_{k=1}^K N_k \left\{ d \ln(2\pi) + \ln |\mathbf{F}_k^{-1}| + \text{tr} \left[\mathbf{F}_k \left(\frac{1}{N_k} \sum_{i=1}^{N_k} \mathbf{x}_{ki} \mathbf{x}_{ki}^T \right) \right] \right\} \quad (14)$$

where $\mathbf{x}_{ki}$ represents the i th data from class k , and N_k is the data size for class k . According to the model definition, the latent feature $\mathbf{y}_{ki}$ satisfies the conditional distribution $p(\mathbf{y}_{ki}|\mathbf{x}_{ki}) = N(\mathbf{y}_{ki}; \mathbf{A}^T \mathbf{F}_k \mathbf{x}_{ki}, \mathbf{I}_m - \mathbf{A}^T \mathbf{F}_k \mathbf{A})$. Given $\mathbf{x}_{ki}$, the latent feature $\mathbf{y}_{ki}$ can be approximated with a conditional mean $\mathbf{y}_{ki} = \mathbf{A}^T \mathbf{F}_k \mathbf{x}_{ki}$. Then the posterior distribution $q(\mathbf{y}|k) = N(\mathbf{y}; \boldsymbol{\mu}_{yk}, \boldsymbol{\Sigma}_{yk})$ with

$$\boldsymbol{\mu}_{yk} = \mathbf{A}^T \mathbf{F}_k \boldsymbol{\mu}_{xk}, \quad \boldsymbol{\Sigma}_{yk} = \mathbf{A}^T \mathbf{F}_k \boldsymbol{\Sigma}_{xk} \mathbf{F}_k \mathbf{A} \quad (15)$$

Then $q(\mathbf{y}) = \sum_{k=1}^K w_k p(\mathbf{y}|k)$.

Similar to the IDA [7], we approximate the entropy of $q(\mathbf{y})$ with a global Gaussian entropy. Thus the MI under the linear statistical model can be expressed as

$$I_{\mathbf{A}, \boldsymbol{\Psi}}(\mathbf{C}, \mathbf{Y}) = H(\mathbf{Y}) - \sum_{k=1}^K w_k H(\mathbf{Y}|k) \\ = \frac{1}{2} \left[\log |\mathbf{A}^T \boldsymbol{\Omega} \mathbf{A}| - \sum_{k=1}^K w_k \log |\mathbf{A}^T \boldsymbol{\Omega}_k \mathbf{A}| \right] \quad (16)$$

where $\boldsymbol{\Omega} = \sum_{k=1}^K w_k [\boldsymbol{\Omega}_k + (\boldsymbol{\mu}_k - \boldsymbol{\mu})(\boldsymbol{\mu}_k - \boldsymbol{\mu})^T]$, $\boldsymbol{\mu} = \sum_{k=1}^K w_k \boldsymbol{\mu}_k$, $\boldsymbol{\mu}_k = \mathbf{F}_k \boldsymbol{\mu}_{xk}$, and $\boldsymbol{\Omega}_k = \mathbf{F}_k \boldsymbol{\Sigma}_{xk} \mathbf{F}_k$. $\boldsymbol{\mu}_{xk}$ and $\boldsymbol{\Sigma}_{xk}$ are the prior mean and the prior covariance matrix for the original data from class k .

3.2. Optimization algorithm

The gradient functions of Eqs. (14) and (16) with respect to parameters $\{\mathbf{A}, \boldsymbol{\Psi}\}$, which is the key point to implement the optimization algorithm, can be derived according to the following theorems. The detailed proofs are provided in the Appendix.

Theorem 1. Let $\text{diag}(\mathbf{S})$ represent a vector with the diagonal elements in the matrix $\mathbf{S}$. According to Eq. (14), the gradient of log-likelihood function $L_{\mathbf{A}, \boldsymbol{\Psi}}(\mathbf{X})$ with respect to the noise variance $\boldsymbol{\Psi}_k$ can be expressed as.

$$\text{diag} \left(\frac{\partial L_{\mathbf{A}, \boldsymbol{\Psi}}(\mathbf{X})}{\partial \boldsymbol{\Psi}_k} \right) = \text{diag} \left(\frac{N_k}{2} \left[\mathbf{F}_k \left(\frac{1}{N_k} \sum_{i=1}^{N_k} \mathbf{x}_{ki} \mathbf{x}_{ki}^T \right) \mathbf{F}_k - \mathbf{F}_k \right] \right) \quad (17)$$

Theorem 2. According to Eq. (16), the gradient of MI function $I_{\mathbf{A}, \boldsymbol{\Psi}}(\mathbf{C}, \mathbf{Y})$ with respect to the noise variance $\boldsymbol{\Psi}_k$ can be expressed as

$$\text{diag} \left(\frac{\partial I_{\mathbf{A}, \boldsymbol{\Psi}}(\mathbf{C}, \mathbf{Y})}{\partial \boldsymbol{\Psi}_k} \right) = \text{diag} \left(w_k \mathbf{F}_k [\mathbf{O}(\boldsymbol{\mu} - \boldsymbol{\mu}_k) \boldsymbol{\mu}_k^T - (\mathbf{O} - \mathbf{O}_k) \boldsymbol{\Omega}_k] \right) \quad (18)$$

where $\mathbf{O} = \mathbf{A}(\mathbf{A}^T \boldsymbol{\Omega} \mathbf{A})^{-1} \mathbf{A}^T$, and $\mathbf{O}_k = \mathbf{A}(\mathbf{A}^T \boldsymbol{\Omega}_k \mathbf{A})^{-1} \mathbf{A}^T$.

Theorem 3. According to Eq. (14), the gradient of log-likelihood function $L_{\mathbf{A}, \boldsymbol{\Psi}}(\mathbf{X})$ with respect to the transformation matrix $\mathbf{A}$ can be expressed as

$$\frac{\partial L_{\mathbf{A}, \boldsymbol{\Psi}}(\mathbf{X})}{\partial \mathbf{A}} = \sum_{k=1}^K N_k \left[\mathbf{F}_k \left(\frac{1}{N_k} \sum_{i=1}^{N_k} \mathbf{x}_{ki} \mathbf{x}_{ki}^T \right) \mathbf{F}_k - \mathbf{F}_k \right] \mathbf{A} \quad (19)$$

Theorem 4. According to Eq. (14), the gradient of MI function $I_{\mathbf{A}, \boldsymbol{\Psi}}(\mathbf{C}, \mathbf{Y})$ with respect to the transformation matrix $\mathbf{A}$ can be expressed as

$$\frac{\partial I_{\mathbf{A}, \boldsymbol{\Psi}}(\mathbf{C}, \mathbf{Y})}{\partial \mathbf{A}} = \boldsymbol{\Omega} \mathbf{A} (\mathbf{A}^T \boldsymbol{\Omega} \mathbf{A})^{-1} - \sum_{k=1}^K w_k \boldsymbol{\Omega}_k \mathbf{A} (\mathbf{A}^T \boldsymbol{\Omega}_k \mathbf{A})^{-1} \\ + \sum_{k=1}^K w_k (\mathbf{P}_k + \mathbf{P}_k^T - \mathbf{M}_k - \mathbf{M}_k^T) \mathbf{A} \quad (20)$$

where $\mathbf{P}_k = \mathbf{F}_k \mathbf{O} (\boldsymbol{\mu} - \boldsymbol{\mu}_k) \boldsymbol{\mu}_k^T$, and $\mathbf{M}_k = \mathbf{F}_k (\mathbf{O} - \mathbf{O}_k) \boldsymbol{\Omega}_k$.

Algorithm 1. Update model parameters $\{\mathbf{A}, \boldsymbol{\Psi}\}$ based on gradient descent method.

Input: training dataset, subspace dimension m .

Output: the optimal transformation matrix $\mathbf{A}^*$, variance matrices $\boldsymbol{\Psi}^* = \{\boldsymbol{\Psi}_k^*\}_{k=1}^K$.

Initialization: $\text{iter} = 0$, maxit , threshold , $\mathbf{A}_0$, $\boldsymbol{\Psi}_0$, λ , and η .

While ($\text{iter} < \text{maxit}$ & $\text{diff} > \text{threshold}$)

$\text{iter} = \text{iter} + 1$;

$\nabla \boldsymbol{\Psi}_k = \frac{\partial I_{\mathbf{A}, \boldsymbol{\Psi}}(\mathbf{C}, \mathbf{Y})}{\partial \boldsymbol{\Psi}_k} + \lambda \frac{\partial L_{\mathbf{A}, \boldsymbol{\Psi}}(\mathbf{X})}{\partial \boldsymbol{\Psi}_k}$; $\boldsymbol{\Psi}_k^{\text{new}} = \boldsymbol{\Psi}_k^{\text{old}} + \eta \nabla \boldsymbol{\Psi}_k$;

$\nabla \mathbf{A} = \frac{\partial I_{\mathbf{A}, \boldsymbol{\Psi}}(\mathbf{C}, \mathbf{Y})}{\partial \mathbf{A}} + \lambda \frac{\partial L_{\mathbf{A}, \boldsymbol{\Psi}}(\mathbf{X})}{\partial \mathbf{A}}$; $\mathbf{A}^{\text{new}} = \mathbf{A}^{\text{old}} + \eta \nabla \mathbf{A}$;

$LF(\text{iter}) = I_{\mathbf{A}^{\text{new}}, \boldsymbol{\Psi}^{\text{new}}}(\mathbf{C}, \mathbf{Y}) + \lambda L_{\mathbf{A}^{\text{new}}, \boldsymbol{\Psi}^{\text{new}}}(\mathbf{X})$;

$\text{diff} = \left| \frac{LF(\text{iter}) - LF(\text{iter} - 1)}{LF(\text{iter} - 1)} \right|$;

 if ($LF(\text{iter}) < LF(\text{iter} - 1)$)

$\eta = \eta/2$;

 end

End

According to the above theorems, the optimization problem in Eq. (13) can be solved directly via gradient descent. The alternating iteration procedure is summarized in Algorithm 1, which starts from a set of initial values. In the following Algorithm 1, iter represents the number of iterations, maxit represents the maximum number of iterations, threshold denotes the convergence threshold (in the following experiments, we set $\text{threshold} = 10^{-3}$), λ is the weighting factor, η is the step size, $\mathbf{A}_0$ denotes the initial transformation matrix, $\boldsymbol{\Psi}_0$ denotes the initial variance.

3.3. Classification and prediction

After the optimization algorithm converges, we can obtain the low-dimensional feature $\mathbf{y}$ under the optimal model parameters $\{\mathbf{A}^*, \boldsymbol{\Psi}^*\}$, and further estimate the posterior distribution $q(\mathbf{y}|k)$ in the training stage. For a test data $\mathbf{x}^*$, we first calculate the low-dimensional feature $\{\mathbf{y}_k^*, \mathbf{y}_k^* = \mathbf{A}^{*T} (\mathbf{A}^* \mathbf{A}^{*T} + \boldsymbol{\Psi}_k^*)^{-1} \mathbf{x}^*\}_{k=1}^K$ under each linear statistical model, then classify the test data via the Bayesian classifier as

$$t^* = \arg \max_{k: 1, 2, \dots, K} q(\mathbf{y}_k^* | k) w_k \quad (21)$$

So far, we implement the entire supervised dimensionality reduction model, and the model parameters are also estimated. In practice, the classification task can be performed in two stages: training and test stages. The specific classification block diagram is shown in Fig. 1.

4. Related methods

In this section, we will compare some existing dimensionality reduction methods with our proposed method, EIDA, of which FA and PCA are unsupervised dimensionality reduction methods, i.e., the feature projection matrix design does not include data labels; LDA, MMLDA and MIM, similar to LDA and EIDA, are supervised.

4.1. FA

In FA [19], the generative model is given by $\mathbf{x} = \mathbf{A}\mathbf{y} + \boldsymbol{\varepsilon}$, where $\mathbf{x} \in \mathbb{R}^{d \times 1}$ represents the original data, $\mathbf{A} \in \mathbb{R}^{d \times m}$ is known as the factor loading matrix, $\mathbf{y} \in \mathbb{R}^{m \times 1}$ represents the low-dimensional factor, and $\boldsymbol{\varepsilon} \in \mathbb{R}^{d \times 1}$ is the noise. Here the factor $\mathbf{y}$ is assumed to be $p(\mathbf{y}) = N(\mathbf{y}; \mathbf{0}, \mathbf{I})$ distributed, the noise $\boldsymbol{\varepsilon}$ is distributed $p(\boldsymbol{\varepsilon}) = N(\boldsymbol{\varepsilon}; \mathbf{0}, \boldsymbol{\Psi})$ with $\boldsymbol{\Psi}$ being a diagonal matrix, and the optimal model parameters $\{\mathbf{A}^*, \boldsymbol{\Psi}^*\}$ can be estimated via maximum likelihood. Given $\{\mathbf{A}^*, \boldsymbol{\Psi}^*\}$, the expected value of the factor can be computed through the linear projection $\mathbf{y} = \mathbf{A}^{*T}(\mathbf{A}^* \mathbf{A}^{*T} + \boldsymbol{\Psi}^*)^{-1} \mathbf{x}$. Similarly, we utilize the training factors to design a Bayesian classifier. For a test data $\mathbf{x}^*$, we first calculate the corresponding factor $\mathbf{y}^*$, then classify the test data via the Bayesian classifier.

4.2. PCA

PCA [4,20], which is generally used for unsupervised dimensionality reduction, can also be used for classification. In PCA model, the transformed data can be expressed as $\mathbf{y} = \boldsymbol{\Phi}\mathbf{x}$, where $\mathbf{x}$ represents the observed data, $\boldsymbol{\Phi}$ denotes the transformation matrix, which is given by the m dominant eigenvectors (i.e., those with the largest associated eigenvalues) of the covariance matrix of the observed data. Similar to FA, a Bayesian classifier can be designed based on the transformed data in the training stage.

4.3. LDA

The optimization function of LDA [5,6] is $J_{LDA} = |\boldsymbol{\Phi}\boldsymbol{\Sigma}_B\boldsymbol{\Phi}^T|/|\boldsymbol{\Phi}\boldsymbol{\Sigma}_W\boldsymbol{\Phi}^T|$, where $\boldsymbol{\Sigma}_B$ and $\boldsymbol{\Sigma}_W$ represent the between-class scattering matrix and the within-class scattering matrix respectively:

$$\boldsymbol{\Sigma}_B = \frac{1}{\sum_{k=1}^K N_k} \sum_k \sum_{j=1, j \neq k}^{N_j} (\mathbf{x}_{ji} - \mathbf{u}_{xk})(\mathbf{x}_{ji} - \mathbf{u}_{xk})^T \quad (22)$$

$$\boldsymbol{\Sigma}_W = \frac{1}{\sum_{k=1}^K N_k} \sum_k \sum_{i=1}^{N_k} (\mathbf{x}_{ki} - \mathbf{u}_{xk})(\mathbf{x}_{ki} - \mathbf{u}_{xk})^T \quad (23)$$

The optimal transformation matrix $\boldsymbol{\Phi}^*$ can be found analytically by maximizing J_{LDA} . Due to $\boldsymbol{\Sigma}_B$ is singular for the dimension of the projection subspace m larger than $K - 1$, m has to be smaller than the number of classes K .

4.4. MMLDA

MMLDA is an improved version of LDA. To overcome the problem of outlier, the median mean line is introduced in MMLDA. For a sample $\mathbf{x}^*$, the median mean line of class k is the line through the median sample $\hat{\mathbf{x}}_k^*$ and the mean vector $\mathbf{u}_{xk}$. The distance from $\hat{\mathbf{x}}_k^*$ to $\mathbf{x}^*$ locates medially in the distance of all the sample from class k to $\mathbf{x}^*$. Due to the median sample $\hat{\mathbf{x}}_k^*$ is nearly stable for the outliers, the median mean line is more robust than the mean vector $\mathbf{u}_{xk}$ to outliers in LDA. The optimization function of MMLDA is $J_{MMLDA} = |\boldsymbol{\Phi}\hat{\boldsymbol{\Sigma}}_B\boldsymbol{\Phi}^T|/|\boldsymbol{\Phi}\hat{\boldsymbol{\Sigma}}_W\boldsymbol{\Phi}^T|$, which is similar to that in LDA. Here the between-class scattering matrix $\hat{\boldsymbol{\Sigma}}_B$ and the within-class scattering matrix $\hat{\boldsymbol{\Sigma}}_W$ are defined to be:

$$\hat{\boldsymbol{\Sigma}}_B = \frac{1}{\sum_{k=1}^K N_k} \sum_k \sum_{j=1, j \neq k}^{N_j} (\mathbf{x}_{ji} - \hat{\mathbf{x}}_{ji}^k)(\mathbf{x}_{ji} - \hat{\mathbf{x}}_{ji}^k)^T \quad (24)$$

$$\hat{\boldsymbol{\Sigma}}_W = \frac{1}{\sum_{k=1}^K N_k} \sum_k \sum_{i=1}^{N_k} (\mathbf{x}_{ki} - \hat{\mathbf{x}}_{ki}^k)(\mathbf{x}_{ki} - \hat{\mathbf{x}}_{ki}^k)^T \quad (25)$$

where K is the number of classes and N_k is the number of training sample of class k . $\hat{\mathbf{x}}_{ij}^k$ is projection of the j th sample from the i th class $\mathbf{x}_{ij}$ onto the k th class's median-mean line [34]. Since $\hat{\boldsymbol{\Sigma}}_B$ in (24) is non-singular for $m = 1, 2, \dots, d$ with m denoting the dimension of the projection subspace and d denoting the dimension of original data, m is not limited to be smaller than $K - 1$ any more in the MMLDA method.

4.5. MIM and its extension algorithm

In MIM [14], the linear transformation function is modeled as $\mathbf{y} = \boldsymbol{\Phi}\mathbf{x} + \boldsymbol{\varepsilon}$, where the noise $\boldsymbol{\varepsilon}$ is distributed $p(\boldsymbol{\varepsilon}) = N(\boldsymbol{\varepsilon}; \mathbf{0}, \mathbf{R}_0^{-1})$ and $\mathbf{R}_0$ is the known noise precision matrix. Because the MI is not easy to calculate, MIM gives an explicit expression for the gradient of Shannon MI with respect to the transformation matrix

$$\frac{\partial I_{MIM}(C; \mathbf{Y})}{\partial \boldsymbol{\Phi}} = \mathbf{R}_0 \boldsymbol{\Phi} \hat{\boldsymbol{\Sigma}} = \mathbf{R}_0 \boldsymbol{\Phi} \sum_{k=1}^K w_k \int p(\mathbf{y}|k) (\mathbf{x}_y(k) - \mathbf{x}_y) (\mathbf{x}_y(k) - \mathbf{x}_y)^T d\mathbf{y} \quad (26)$$

where $\mathbf{x}_y(k) = \int \mathbf{x} p(\mathbf{x}|\mathbf{y}, k) d\mathbf{x}$ and $\mathbf{x}_y = \int \mathbf{x} p(\mathbf{x}|\mathbf{y}) d\mathbf{x}$. $\hat{\boldsymbol{\Sigma}}$ is known as the equivalent MMSE matrix, which can be computed via Monte Carlo simulation. Then the optimal transformation matrix $\boldsymbol{\Phi}^*$ can be solved directly using gradient descent. In summary, MIM

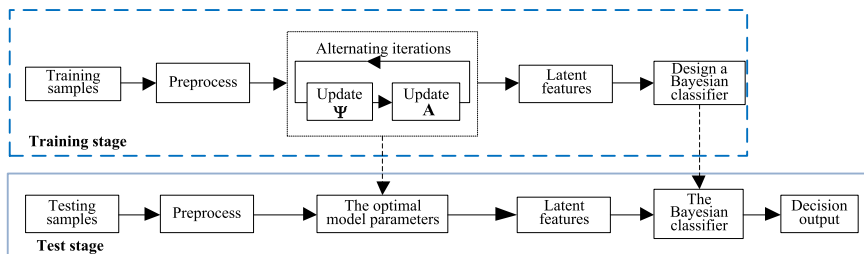


Fig. 1. The classification block diagram of the proposed method

estimates the first and second order statistics of the data in the high-dimensional space, and then designs the transformation matrix based on the information-theoretic criteria. However, in practice, especially in the case of limited data, it is difficult to accurately estimate these statistics in the original high-dimensional space. Thus the transformation matrix obtained may be non-optimal, which will affect the final classification performance.

Different from MIM, our method, EIDA, describes the statistical structure of the transformed low-dimensional subspace via a linear statistical model, and utilizes the MI based information-theoretic criterion to further maximizes the between-class separability. In EIDA, we directly estimate the statistics of the transformed features in the low-dimensional subspace. Table 2 depicts the numbers of parameters to be estimated in EIDA and MIM. In EIDA, we need to estimate $\{\mathbf{A}, \Psi = \{\psi_k\}_{k=1}^K\}$ with the constraint $\max_{\mathbf{A}, \Psi} I_{\mathbf{A}, \Psi}(C, \mathbf{Y}) + \lambda L_{\mathbf{A}, \Psi}(\mathbf{X})$. The upper bound of the number of parameters to be estimated for $\mathbf{A}$ is $dm - 0.5m(m - 1)$, and the number of parameters to be estimated for $\{\Psi = \{\psi_k\}_{k=1}^K\}$ is Kd . Thus the total number of parameters to be estimated for EIDA is $dm - 0.5m(m - 1) + Kd$, as shown in Table 2. In MIM, we need to estimate $\{\{\mu_{xk}, \Sigma_{xk}\}_{k=1}^K, \Phi\}$ for MIM. The numbers of parameters to be estimated for $\{\{\mu_{xk}\}_{k=1}^K\}$ and $\{\{\Sigma_{xk}\}_{k=1}^K\}$ are Kd and $0.5K(d + 1)d$, the number of parameters to be estimated for Φ is $dm - 0.5m(m - 1)$. Thus the total number of parameters to be estimated for MIM is $K(d + 0.5(d + 1)d) + dm - 0.5m(m - 1)$, as shown in Table 2. In the above expressions, K denotes the number of classes, d denotes the original data dimension and m denotes the size of subspace. Obviously, for the given K , d and m , the number of parameters to be estimated in EIDA is smaller than that in MIM.

To decrease the number of parameters to be estimated in MIM, we can also use the parameter estimation method proposed in [31], which estimates the signal statistics under the assumption of intrinsic low-dimensional signals via Mixture of Factor Analyzers (MFA) model. Due to the low-dimensional representation in MFA, [31] can achieve higher parameter estimation accuracy than the GMM parameter estimation in the original high-dimensional data space with a smaller number of data. We can first estimate the statistics in the original high-dimensional space via [31] for each class, and then maximize the MIM objective with the known statistics.

In the above framework, due to the different projections, the low-dimensional latent variable in MFA and the low-dimensional feature in MIM are different and learned separately; while in our method, we assume the same the low-dimensional latent feature in both the linear statistical model $\mathbf{x} = \mathbf{A}\mathbf{y} + \epsilon$ and the MI function $I_{\mathbf{A}, \Psi}(C, \mathbf{Y})$, and maximize not only the log-likelihood function of the linear statistical model but also the separability between the latent features from different classes via joint optimization $\max_{\mathbf{A}, \Psi} I_{\mathbf{A}, \Psi}(C, \mathbf{Y}) + \lambda L_{\mathbf{A}, \Psi}(\mathbf{X})$. As discussed in Section 2.1, in Bayesian classifiers, the class-conditional PDF of the extracted low-dimensional feature is directly used for classification. Compared with the

separate learning framework, our method can directly optimize the class-conditional PDF of the low-dimensional feature $p(\mathbf{y}|k)$ via the maximization of $L_{\mathbf{A}, \Psi}(\mathbf{X})$.

In addition, in the above framework, since the statistics of each class are learned separately, we will use the different loading matrixes $\{\mathbf{A}_k\}_{k=1}^K$ and the different noise variance matrixes $\{\psi_k\}_{k=1}^K$ for different classes; while in our method, the transformation matrix $\mathbf{A}$ is shared for all classes, and the noise variance matrixes $\{\psi_k\}_{k=1}^K$ are different for the different classes. Therefore, the unknown parameters in our method are fewer than those in the above framework. The reason why we share $\mathbf{A}$ in our linear statistical model is to seek a transformation matrix $\mathbf{A}$ such that the latent feature $\mathbf{y}$ is the most informative in identifying the underlying class label c , of which the framework is similar to the IDA method [7]. According to the idea of multi-task learning (MTL) [32], effective parameter sharing among different learning tasks may improve the parameter estimation accuracy with smaller number of data.

5. Experiments and results

5.1. Visual results on synthetic datasets

Some visual results of the proposed method on synthetic datasets are presented in this subsection, in order to examine whether the proposed method can combine linear statistical model with information theory effectively.

Firstly, we simulate two classes of single Gaussian data in the 2-dimensional plane with 200 samples in each class, and set $\lambda = 0.001$, $\eta = 0.01$. The synthetic dataset and experimental results are shown in Fig. 2.

In Fig. 2(a), the circles and the crosses are the single Gaussian data points in the original space. Fig. 2(b)–(d) respectively show the 1-dimensional features learned via EIDA, FA and LDA methods on the single Gaussian data. It is clear to see that EIDA obtains the maximum separable 1-dimensional features from the two classes of single Gaussian data, as shown in Fig. 2(b). To quantitatively compare these three feature extraction methods, Table 3 further gives the classification accuracies via linear Bayesian classifiers on the synthetic single Gaussian dataset. Obviously, the proposed method, EIDA, yields the highest accuracies on the single Gaussian data.

Secondly, we simulate two classes of GMM data in the 2-dimensional plane with 200 samples in each class, and set $\lambda = 0.001$, $\eta = 0.01$, to evaluate the performance of the proposed method on non-Gaussian datasets. The synthetic dataset and experimental results are shown in Fig. 3.

Fig. 3(a) shows the GMM data points in the original space. Fig. 3(b)–(d) respectively show the 1-dimensional features learned via EIDA, FA and LDA methods on the GMM data. It is clear to see that EIDA obtains the maximum separable 1-dimensional features from the two classes of GMM data, as shown in Fig. 3(b). To quantitatively compare these three feature extraction methods, Table 4 further gives the classification accuracies via linear Bayesian classifiers on the synthetic non-Gaussian dataset. Obviously, the proposed method, EIDA, yields the highest accuracies on the GMM data.

These two examples demonstrate the effectiveness of the proposed method, which designs the transformation matrix via the joint optimization of MI function and log-likelihood function.

5.2. Introduction of benchmark datasets

The performance of the proposed method is evaluated

Table 2
Numbers of parameters to be estimated in MIM and EIDA.

Method	Numbers of parameters to be estimated (K : the number of classes, d : the original data dimension, m : the size of subspace)
MIM	$K(d + 0.5(d + 1)d) + dm - 0.5m(m - 1)$
EIDA	$\leq dm - 0.5m(m - 1) + Kd$

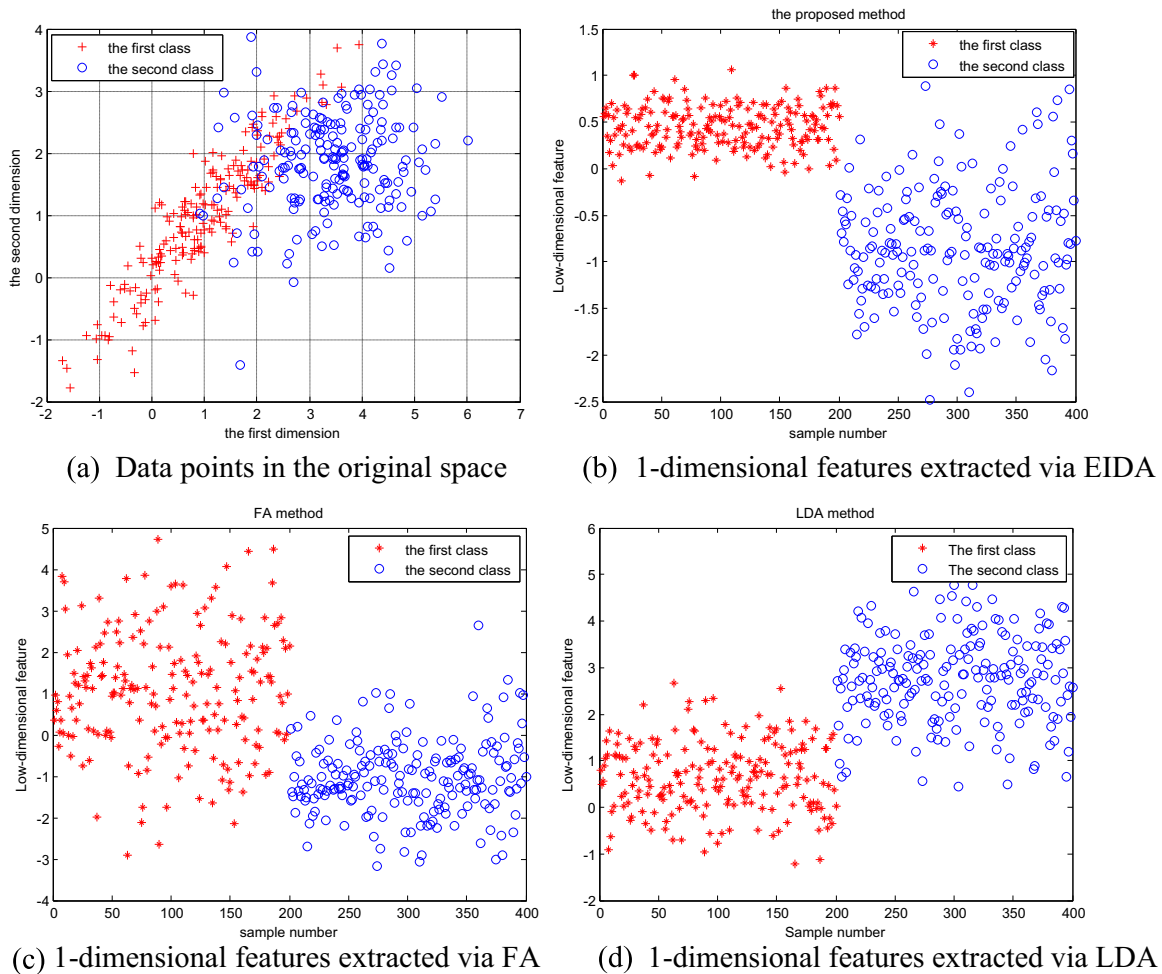


Fig. 2. Some results on the first synthetic dataset.

Table 3

The classification accuracies of EIDA, FA and LDA learned features on the first synthetic dataset.

Method	EIDA	FA	LDA
Classification accuracy	0.9150	0.8300	0.8975

experimentally with six benchmark datasets: Heart, Satellite, Breast_cancer, Ringnorm, Waveform and Twonorm in the following.¹ The datasets consist of various number of classes and attributes with different data sizes, a detailed description is summarized in Table 5.

For the Satellite dataset, training and test sets have been specifically designed by its donor, of which there are 4435 training data and 2000 test data in the Satellite dataset. For all other datasets, we randomly split the dataset into two non-overlapping groups of equal size, then select one group as training set and the other one as test set. Therefore, the model parameters can be estimated from the training set and validated on the test set. To control the variability caused by random partitions, the splitting work is repeated 20 times and the results are averaged over the 20 partitions.

As shown in Table 2, the number of parameters to be estimated

in EIDA is smaller than that in MIM for a given dataset. Fig. 4 depicts the numbers of parameters to be estimated for the Satellite dataset introduced in Table 5 via EIDA and MIM, where the upper bounds are given for EIDA.

5.3. Classification accuracies on benchmark datasets

Five related methods are considered to compare against the proposed method: FA, PCA, LDA, IDA, MIM. The performance is evaluated using linear (L) and quadratic (Q) Bayesian classifiers, with prior probabilities estimated empirically from the data and the Gaussian approximation of the class-conditional PDFs. The subspace dimensions are $1 \leq m \leq d$ for FA, PCA, MMLDA, IDA, MIM and the proposed method, $1 \leq m \leq K - 1$ for LDA. The step sizes of the gradient descent in IDA, MIM and the proposed method are set to be 0.01. The experimental results are summarized in the following Tables 6–11, where the best performance for each m , i.e., a given size of subspace, is shown in bold and the best overall performance is shown in bold italic. The performances, represented by the classification accuracies and standard deviations, are shown in Tables 6–11 for various datasets, of which Tables 6, 8–11 respectively show the average classification accuracies and standard derivations for the Heart, Breast_cancer, Ringnorm, Waveform and Twonorm datasets over the 20 random partitions for the training and test sets; and Table 7 shows the classification accuracies for the Satellite dataset on the fixed partition specifically designed by its donor. In Tables 6–11, LDA cannot work when $m \geq K$. We observe that the best performance in each column, i.e., for each m , is

¹ The datasets Heart, Satellite, Breast_cancer, Ringnorm, Waveform and Twonorm can be downloaded from the UCI Machine Learning Repository (<http://archive.ics.uci.edu/ml/datasets.html>).

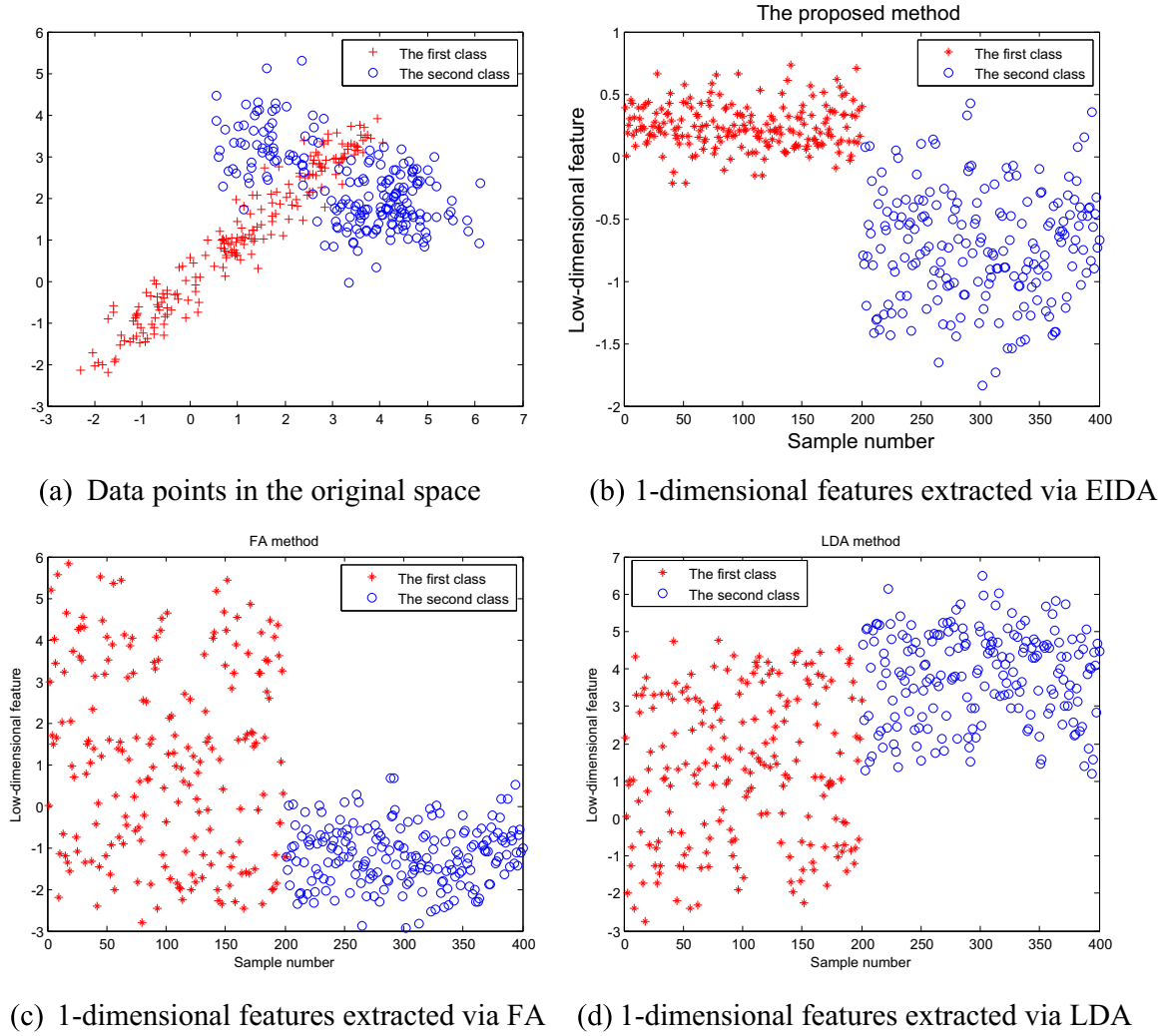


Fig. 3. Some results on the second synthetic dataset.

Table 4

The classification accuracies of EIDA, FA and LDA learned features on the second synthetic dataset.

Method	EIDA	FA	LDA
Classification accuracy	0.9525	0.8225	0.7350

Table 5

Description of the six datasets.

Dataset	Number of classes (K)	Data dimension (d)	Data size (n)
Heart	2	13	270
Satellite	6	36	4435/2000
Breast_cancer	2	9	277
Ringnorm	2	20	7400
Waveform	3	21	5000
Twonorm	2	20	7400

obtained by the proposed method in most cases.

According to Table 5, the size-dimension ratios, i.e., n/d , of each dataset are 20, 178, 30, 370, 238, and 370. Obviously, the ratios of the last three datasets are much larger than those of the first three, which means the parameter estimation tasks for the first three datasets are much harder than the last three datasets. As shown in Tables 6–11, for the last three datasets, although the best results

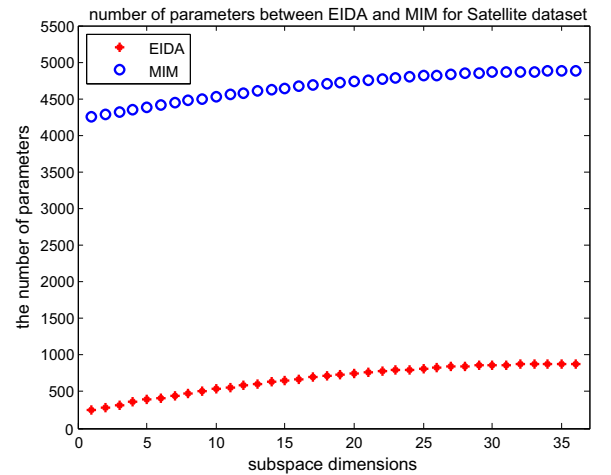


Fig. 4. The numbers of parameters to be estimated for the Satellite dataset introduced in Table 5 via EIDA and MIM, where the upper bounds are given for EIDA. The calculation formulas are depicted in Table 2.

are obtained by the proposed method, its classification accuracies are just slightly better than those of other methods; however, for the first three datasets, the proposed method significantly outperforms other methods. This accords with our analysis in Section

Table 6

Average classification accuracies (%) and standard deviations (%) for Heart data over the 20 random partitions for the training and test sets.

Method		Size of subspace (m)				
		1	2	3	4	5
FA	L	79.37 $\pm$ 1.80	79.37 $\pm$ 2.68	80.67 $\pm$ 2.27	81.22 $\pm$ 2.42	82.44 $\pm$ 1.94
	Q	79.26 $\pm$ 1.88	79.89 $\pm$ 2.72	80.63 $\pm$ 2.96	80.37 $\pm$ 2.49	81.26 $\pm$ 1.92
PCA	L	56.67 $\pm$ 3.04	68.19 $\pm$ 3.20	68.59 $\pm$ 3.44	68.26 $\pm$ 3.91	78.59 $\pm$ 2.15
	Q	56.44 $\pm$ 4.08	67.22 $\pm$ 3.51	66.85 $\pm$ 3.19	65.70 $\pm$ 4.01	76.22 $\pm$ 2.56
LDA	L	83.22 $\pm$ 1.93	–	–	–	–
	Q	83.19 $\pm$ 2.07	–	–	–	–
MMLDA	L	82.89 $\pm$ 2.20	83.33 $\pm$ 2.48	83.19 $\pm$ 1.94	83.11 $\pm$ 1.92	83.22 $\pm$ 1.82
	Q	83.04 $\pm$ 2.33	82.78 $\pm$ 2.52	82.41 $\pm$ 2.19	82.30 $\pm$ 2.61	81.67 $\pm$ 2.62
IDA	L	75.78 $\pm$ 3.69	82.67 $\pm$ 3.79	83.85 $\pm$ 2.62	83.59 $\pm$ 2.15	83.44 $\pm$ 2.55
	Q	75.67 $\pm$ 3.67	82.33 $\pm$ 3.91	81.78 $\pm$ 3.25	80.48 $\pm$ 2.86	79.70 $\pm$ 3.16
MIM	L	77.93 $\pm$ 4.20	83.19 $\pm$ 1.95	83.30 $\pm$ 1.90	83.07 $\pm$ 2.01	83.37 $\pm$ 2.05
	Q	77.89 $\pm$ 4.33	82.41 $\pm$ 2.32	82.85 $\pm$ 2.36	82.26 $\pm$ 2.00	81.37 $\pm$ 2.70
EIDA	L	84.52 $\pm$ 1.46	84.22 $\pm$ 1.75	84.11 $\pm$ 2.34	84.33 $\pm$ 2.44	83.85 $\pm$ 2.36
	Q	84.78 $\pm$ 1.55	84.70 $\pm$ 1.72	84.22 $\pm$ 2.07	83.93 $\pm$ 2.44	83.41 $\pm$ 2.71

Table 7

Classification accuracies (%) for Satellite data on the fixed partition specifically designed by its donor.

Method		Size of subspace (m)				
		1	3	5	7	8
FA	L	57.75	80.95	80.30	80.95	81.35
	Q	57.80	84.25	83.35	84.25	84.05
PCA	L	50.90	80.50	80.40	81.00	81.25
	Q	51.10	83.70	84.10	83.80	83.75
LDA	L	52.35	82.30	82.85	–	–
	Q	56.50	84.15	84.45	–	–
MMLDA	L	46.55	81.65	81.80	82.25	82.45
	Q	59.55	84.10	84.20	84.35	84.25
IDA	L	59.30	82.00	82.05	81.90	81.80
	Q	67.30	84.70	83.80	83.75	83.85
MIM	L	71.55	81.55	81.20	81.45	81.50
	Q	72.65	83.70	84.10	83.85	83.95
EIDA	L	51.75	82.00	83.25	83.05	82.05
	Q	53.50	85.15	86.20	86.50	85.70

5. When the size-dimension ratio is large, such as the last three datasets, other methods like MIM can estimate the first and second order statistics in the original high-dimensional space accurately, and the proposed method, which designs the transformation matrix and estimates these statistics in the low-dimensional feature space simultaneously, has no significant advantage. When the size-dimension ratio is small, *i.e.*, in the case of high-dimensional limited data, such as the first three datasets, other methods cannot estimate these statistics accurately and their performances are generally weaker. In such cases, not only the estimation errors

in the low-dimensional feature space can be reduced, but also the between-class separability can be improved, the proposed method significantly outperforms other methods.

Comparing the linear classifiers with the quadratic classifiers, the linear classifier is superior to the quadratic classifier on Heart and Waveform, while the quadratic classifier is superior to the linear classifier on Satellite and Ringnorm. Therefore, whether a classifier can perform well on a dataset also depends on the specific characteristics of the dataset.

5.4. Analysis on parameter λ

In our optimization problem in (13), the parameter λ is a critical parameter, which will affect the final performance of our method. When $\lambda = 0$, only the MI objective $I_{A,\Psi}(C, Y)$ will take the effect in (13), and our method is simplified to be IDA; when $\lambda \rightarrow \infty$, only the likelihood objective $L_{A,\Psi}(X)$ will take the effect in (13), and our method is similar to be FA. From the experimental results in Tables 6–11, we can see that the average classification accuracies of our method are higher than those of IDA and FA in most cases. Therefore, our optimal values of λ should be between the range $(0, \infty)$.

Fig. 5(a–f) show the variation of the classification accuracy of our method with the value of parameter λ . As introduced in Section 5.2, there are 20 random partitions for the training and test sets for the Heart, Breast_cancer, Ringnorm, Waveform and Twonorm datasets in our experiment; while there is the fixed partition specifically designed by its donor for the Satellite dataset. In Fig. 5, we randomly give three examples among the different partitions for the Heart, Breast_cancer, Ringnorm, Waveform and Twonorm datasets. From the results in Fig. 5, we can see that our optimal values of λ are between the range $(0, 0.01)$ or $(0, 0.1)$. Fig. 6(a) and (b) show two examples of the variation of the ratio of likelihood value $L_{A,\Psi}(X)$ to MI value $I_{A,\Psi}(C, Y)$ with the iteration in our experiment for the Breast_cancer and Twonorm datasets. The results for other datasets are similar, and we omit these to avoid redundancy. We can see that the ratio value $L_{A,\Psi}(X)/I_{A,\Psi}(C, Y)$ is relatively large here. Thus the optimal λ is relatively small, according to the optimization problem $\max_{A,\Psi} I_{A,\Psi}(C, Y) + \lambda L_{A,\Psi}(X)$. In

Table 8

Average classification accuracies (%) and standard deviations (%) for Breast_cancer data over the 20 random partitions for the training and test sets.

Method		Size of subspace (m)				
		1	2	3	4	5
FA	L	71.47 $\pm$ 3.46	72.34 $\pm$ 2.73	71.80 $\pm$ 3.47	72.34 $\pm$ 3.85	72.52 $\pm$ 3.75
	Q	71.47 $\pm$ 3.50	72.23 $\pm$ 3.05	72.34 $\pm$ 3.12	72.16 $\pm$ 3.11	72.12 $\pm$ 3.34
PCA	L	71.65 $\pm$ 3.30	72.45 $\pm$ 3.13	73.02 $\pm$ 2.83	72.37 $\pm$ 3.22	72.55 $\pm$ 2.93
	Q	71.65 $\pm$ 3.41	72.23 $\pm$ 3.16	73.06 $\pm$ 3.13	73.45 $\pm$ 2.71	72.48 $\pm$ 2.99
LDA	L	72.05 $\pm$ 3.46	–	–	–	–
	Q	71.94 $\pm$ 3.57	–	–	–	–
MMLDA	L	70.90 $\pm$ 3.57	71.37 $\pm$ 3.72	71.40 $\pm$ 3.74	71.29 $\pm$ 3.78	71.26 $\pm$ 3.41
	Q	70.68 $\pm$ 3.33	71.26 $\pm$ 3.73	70.72 $\pm$ 3.44	70.58 $\pm$ 3.78	70.36 $\pm$ 3.48
IDA	L	70.47 $\pm$ 3.30	71.98 $\pm$ 3.00	71.80 $\pm$ 3.47	71.83 $\pm$ 3.52	71.65 $\pm$ 3.52
	Q	70.11 $\pm$ 3.18	72.45 $\pm$ 3.61	72.12 $\pm$ 3.75	71.65 $\pm$ 3.69	71.22 $\pm$ 3.63
MIM	L	70.58 $\pm$ 2.83	72.30 $\pm$ 3.07	71.65 $\pm$ 3.41	71.15 $\pm$ 3.42	71.87 $\pm$ 3.42
	Q	69.86 $\pm$ 2.76	72.48 $\pm$ 3.07	71.65 $\pm$ 3.19	71.47 $\pm$ 3.59	71.47 $\pm$ 3.53
EIDA	L	73.74 $\pm$ 2.83	73.74 $\pm$ 3.14	73.53 $\pm$ 3.40	73.71 $\pm$ 3.42	74.17 $\pm$ 3.01
	Q	73.56 $\pm$ 2.97	73.85 $\pm$ 2.98	73.45 $\pm$ 3.37	73.38 $\pm$ 3.02	73.74 $\pm$ 2.69

Table 9

Average classification accuracies (%) and standard deviations (%) for Ringnorm data over the 20 random partitions for the training and test sets.

Method		Size of subspace (m)				
		1	4	5	7	10
FA	L	76.56 $\pm$ 0.48	76.61 $\pm$ 0.43	76.61 $\pm$ 0.53	76.83 $\pm$ 0.52	76.96 $\pm$ 0.62
	Q	77.01 $\pm$ 0.41	87.02 $\pm$ 0.61	89.12 $\pm$ 0.41	91.95 $\pm$ 0.33	94.89 $\pm$ 0.45
PCA	L	76.66 $\pm$ 0.50	76.79 $\pm$ 0.42	76.85 $\pm$ 0.44	76.90 $\pm$ 0.52	76.93 $\pm$ 0.53
	Q	77.19 $\pm$ 0.41	86.94 $\pm$ 0.61	89.07 $\pm$ 0.41	91.95 $\pm$ 0.39	94.92 $\pm$ 0.33
LDA	L	77.08 $\pm$ 0.59	–	–	–	–
	Q	77.56 $\pm$ 0.48	–	–	–	–
MMLDA	L	77.04 $\pm$ 0.51	77.03 $\pm$ 0.52	77.02 $\pm$ 0.53	77.05 $\pm$ 0.55	77.07 $\pm$ 0.54
	Q	77.57 $\pm$ 0.47	87.12 $\pm$ 0.62	89.10 $\pm$ 0.54	92.02 $\pm$ 0.38	94.95 $\pm$ 0.39
IDA	L	76.83 $\pm$ 0.54	76.88 $\pm$ 0.50	76.91 $\pm$ 0.55	76.97 $\pm$ 0.54	77.04 $\pm$ 0.60
	Q	77.32 $\pm$ 0.53	87.05 $\pm$ 0.52	88.87 $\pm$ 0.51	91.74 $\pm$ 0.46	95.00 $\pm$ 0.36
MIM	L	77.05 $\pm$ 0.55	76.99 $\pm$ 0.53	76.93 $\pm$ 0.49	76.74 $\pm$ 0.52	76.43 $\pm$ 0.47
	Q	77.46 $\pm$ 0.45	87.03 $\pm$ 0.49	88.99 $\pm$ 0.37	91.99 $\pm$ 0.33	95.03 $\pm$ 0.28
EIDA	L	77.24 $\pm$ 0.45	77.50 $\pm$ 0.49	77.49 $\pm$ 0.50	77.56 $\pm$ 0.48	77.49 $\pm$ 0.56
	Q	77.61 $\pm$ 0.45	87.59 $\pm$ 0.51	89.60 $\pm$ 0.38	92.52 $\pm$ 0.38	95.33 $\pm$ 0.30

addition, since the absolute value of the ratios in Fig. 6(b) are much larger than those in Fig. 6(a), the optimal values of λ for the Twonorm dataset are smaller than those for the Breast_cancer dataset, as shown in Fig. 5(c) and (f).

5.5. Robustness evaluation

The aim of this subsection is to investigate the sensitivity of the proposed method to the size of training set, which is a critical variable in the supervised classification.

We first test on a synthetic dataset, which is generated from different Gaussian distribution, with the number of classes $K = 2$, the sample dimension $d = 50$, and the number of total samples $n = 500$ ($n/K = 250$ samples in each class). Fig. 7 shows the variation of the classification accuracy with the ratio of the smallest number of training samples in each class to the sample dimension, where the ratio is denoted as $r = \min_{k=1,2,\dots,K} \{N_k\}/d$ with the number of training samples in the k th class N_k decreasing from $n/K = 250$ with $k = 1, 2$. In the following experiment, the size of subspace $m = 1$ for all dimensionality reduction methods. In Fig. 7(a), we

Table 10

Average classification accuracies (%) and standard deviations (%) for Waveform data over the 20 random partitions for the training and test sets.

Method		Size of subspace (m)				
		1	2	3	4	5
FA	L	58.60 $\pm$ 0.65	86.40 $\pm$ 0.59	86.40 $\pm$ 0.61	86.39 $\pm$ 0.64	86.37 $\pm$ 0.60
	Q	61.22 $\pm$ 0.52	86.13 $\pm$ 0.44	86.19 $\pm$ 0.48	86.13 $\pm$ 0.56	86.05 $\pm$ 0.51
PCA	L	58.59 $\pm$ 0.59	86.51 $\pm$ 0.63	86.43 $\pm$ 0.68	86.42 $\pm$ 0.66	86.31 $\pm$ 0.64
	Q	61.42 $\pm$ 0.48	86.13 $\pm$ 0.46	86.14 $\pm$ 0.50	86.05 $\pm$ 0.50	86.05 $\pm$ 0.45
LDA	L	59.35 $\pm$ 0.72	85.53 $\pm$ 0.63	–	–	–
	Q	61.13 $\pm$ 0.65	85.39 $\pm$ 0.67	–	–	–
MMLDA	L	59.55 $\pm$ 0.79	85.68 $\pm$ 0.62	85.66 $\pm$ 0.59	85.69 $\pm$ 0.59	85.65 $\pm$ 0.67
	Q	61.34 $\pm$ 0.62	85.49 $\pm$ 0.61	85.52 $\pm$ 0.56	85.40 $\pm$ 0.59	85.45 $\pm$ 0.57
IDA	L	58.64 $\pm$ 0.89	86.26 $\pm$ 0.65	86.23 $\pm$ 0.64	86.07 $\pm$ 0.66	85.98 $\pm$ 0.67
	Q	60.33 $\pm$ 0.89	85.98 $\pm$ 0.54	86.01 $\pm$ 0.58	85.68 $\pm$ 0.52	85.65 $\pm$ 0.54
MIM	L	58.96 $\pm$ 0.89	86.16 $\pm$ 0.60	86.17 $\pm$ 0.63	86.03 $\pm$ 0.67	86.00 $\pm$ 0.67
	Q	60.62 $\pm$ 0.73	85.91 $\pm$ 0.53	85.88 $\pm$ 0.61	85.73 $\pm$ 0.64	85.70 $\pm$ 0.57
EIDA	L	61.39 $\pm$ 0.68	86.61 $\pm$ 0.59	86.33 $\pm$ 0.59	85.98 $\pm$ 0.85	85.79 $\pm$ 0.83
	Q	66.44 $\pm$ 0.49	86.42 $\pm$ 0.47	86.35 $\pm$ 0.44	86.29 $\pm$ 0.54	86.16 $\pm$ 0.50

Table 11

Average classification accuracies (%) and standard deviations (%) for Twonorm data over the 20 random partitions for the training and test sets.

Method		Size of subspace (m)				
		1	2	3	4	5
FA	L	97.80 $\pm$ 0.16	97.80 $\pm$ 0.17	97.80 $\pm$ 0.18	97.81 $\pm$ 0.17	97.81 $\pm$ 0.18
	Q	97.80 $\pm$ 0.17	97.80 $\pm$ 0.17	97.80 $\pm$ 0.18	97.80 $\pm$ 0.17	97.79 $\pm$ 0.18
PCA	L	97.80 $\pm$ 0.18	97.80 $\pm$ 0.17	97.79 $\pm$ 0.17	97.80 $\pm$ 0.18	97.81 $\pm$ 0.18
	Q	97.79 $\pm$ 0.19	97.80 $\pm$ 0.17	97.79 $\pm$ 0.17	97.79 $\pm$ 0.17	97.78 $\pm$ 0.19
LDA	L	97.76 $\pm$ 0.21	–	–	–	–
	Q	97.76 $\pm$ 0.18	–	–	–	–
MMLDA	L	97.79 $\pm$ 0.20	97.79 $\pm$ 0.19	97.79 $\pm$ 0.19	97.78 $\pm$ 0.21	97.79 $\pm$ 0.19
	Q	97.78 $\pm$ 0.20	97.78 $\pm$ 0.19	97.77 $\pm$ 0.20	97.77 $\pm$ 0.20	97.76 $\pm$ 0.19
IDA	L	97.77 $\pm$ 0.20	97.76 $\pm$ 0.20	97.76 $\pm$ 0.21	97.76 $\pm$ 0.20	97.76 $\pm$ 0.21
	Q	97.76 $\pm$ 0.19	97.76 $\pm$ 0.20	97.74 $\pm$ 0.21	97.73 $\pm$ 0.20	97.74 $\pm$ 0.21
MIM	L	97.77 $\pm$ 0.20	97.77 $\pm$ 0.20	97.77 $\pm$ 0.20	97.76 $\pm$ 0.20	97.76 $\pm$ 0.20
	Q	97.76 $\pm$ 0.20	97.77 $\pm$ 0.19	97.74 $\pm$ 0.20	97.74 $\pm$ 0.20	97.74 $\pm$ 0.20
EIDA	L	97.85 $\pm$ 0.17	97.87 $\pm$ 0.18	97.89 $\pm$ 0.18	97.90 $\pm$ 0.17	97.89 $\pm$ 0.19
	Q	97.85 $\pm$ 0.17	97.85 $\pm$ 0.18	97.85 $\pm$ 0.18	97.85 $\pm$ 0.19	97.86 $\pm$ 0.20

give the results of all methods, i.e., PCA, FA, LDA, MMLDA, IDA, MIM and EIDA, of which LDA, MMLDA, IDA and MIM cannot work when $r < 1.2$ and FA cannot work when $r < 0.6$. In Fig. 7(b), we only give the results of PCA, FA and EIDA, to show their differences more clearly. It can be seen from Fig. 7, the proposed method, EIDA, outperforms other methods when $r \geq 0.8$ on the synthetic dataset. When $r = 0.6$ on the synthetic dataset, the accuracy of FA is a little better than that of EIDA; when $r = 0.2$ on the synthetic

dataset, the accuracy of PCA is a little better than that of EIDA.

Then we select two benchmark datasets, Ringnorm and Waveform, to evaluate the robustness of the dimensionality reduction methods with small number of training data. Fig. 8 shows the variation of the classification accuracies with r on Ringnorm and Waveform datasets. As shown in Table 5, for Ringnorm data, the number of classes $K = 2$, the sample dimension $d = 20$, the data size $n = 7400$; for Waveform data, the number of classes $K = 3$, the

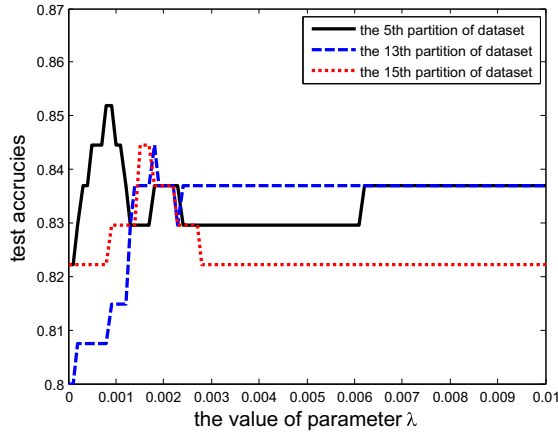
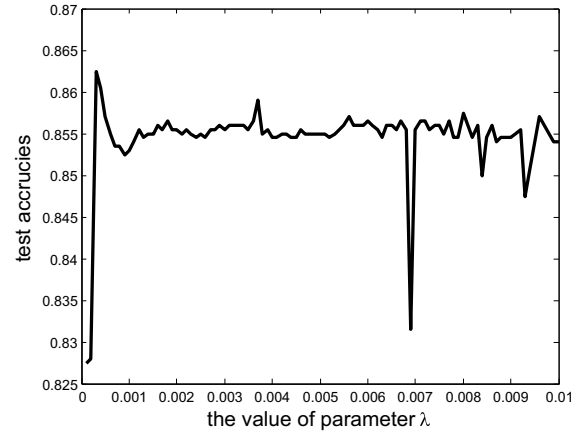
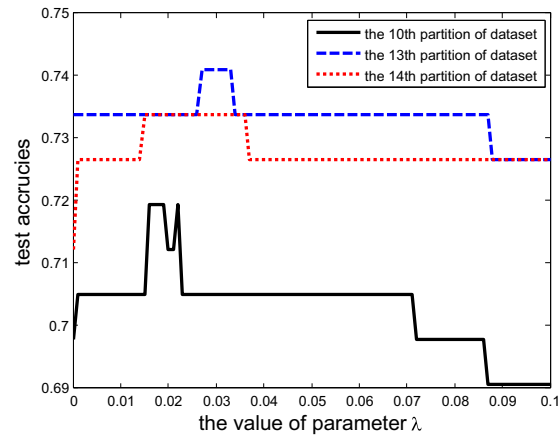
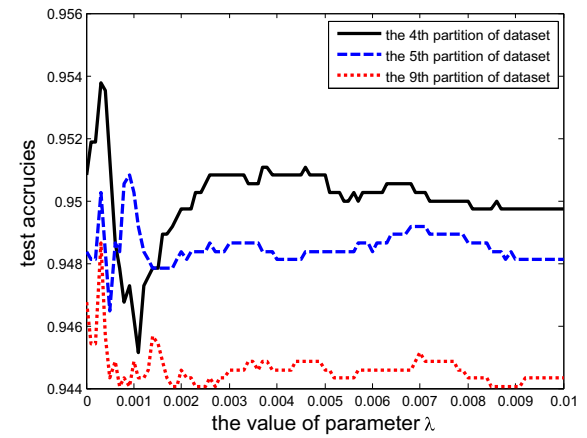
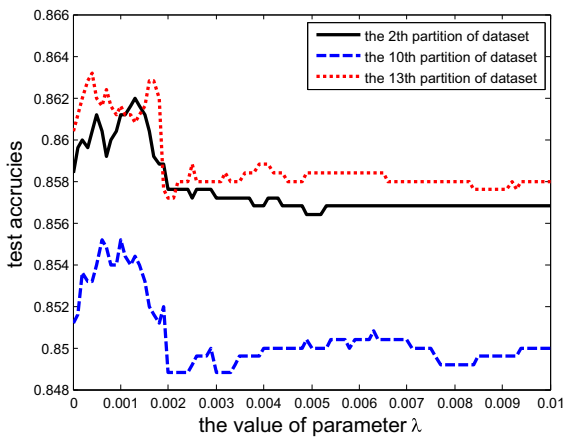
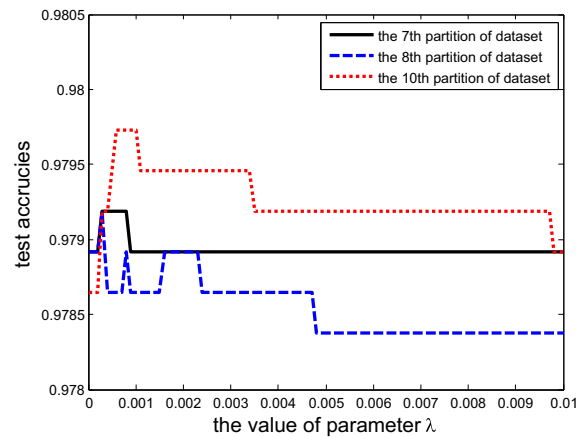
(a) Heart data with $m = 1$ (b) Satellite data with $m = 5$ (c) Breast_cancer data with $m = 1$ (d) Ringnorm data with $m = 10$ (e) Waveform data with $m = 2$ (f) Twonorm data with $m = 2$

Fig. 5. The variation of the classification accuracy of our method with the value of parameter λ . Here m denotes the size of subspace.

sample dimension $d = 21$, the data size $n = 5000$. Here for these two datasets, the training sets are randomly drawn from the dataset with number of samples in the k th class $N_k = r \times d$ with $k = 1, 2, \dots, K$. To control the variability caused by random partitions, the partition work is repeated 50 times, and the results are averaged over the 50 partitions and shown in Fig. 8. Fig. 8(a) and (b) show the variation of the classification accuracies with r on

Ringnorm data, where Fig. 8(a) gives the results with $1.5 \leq r \leq 75$, Fig. 8(b) gives the results with $1.5 \leq r \leq 3$ to show the differences among all methods more clearly. In this experiment on Ringnorm data, we set the size of subspace $m = 2$. Since LDA cannot work when $m \geq K$, we only compare the results of PCA, FA, MMLDA, IDA, MIM and EIDA in Fig. 6(a) and (b). It can be seen from Fig. 8(a) and (b), the proposed method, EIDA, outperforms other methods when

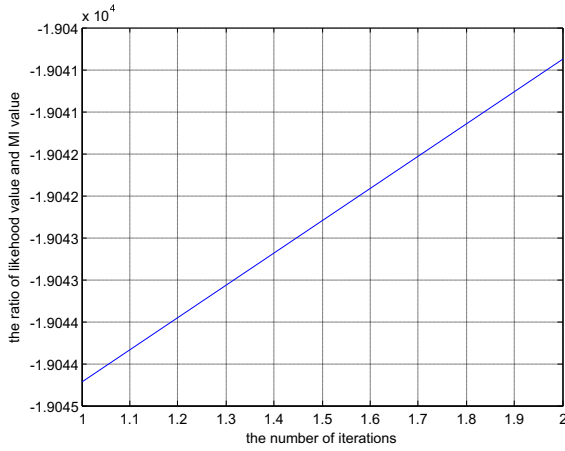
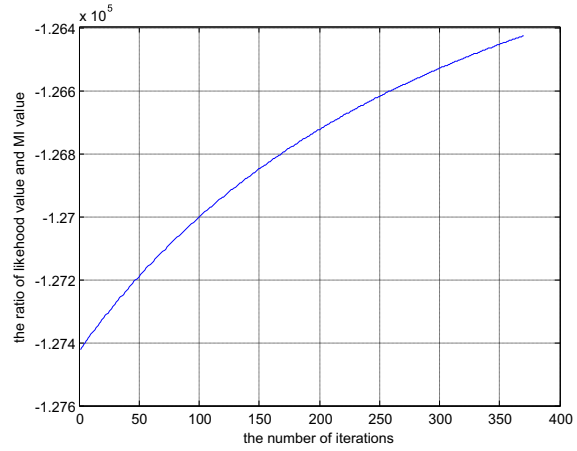
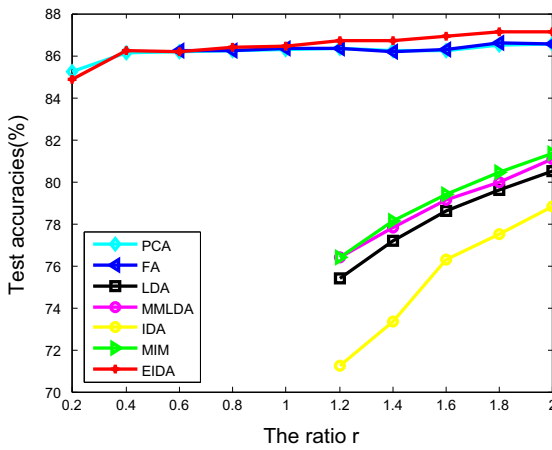
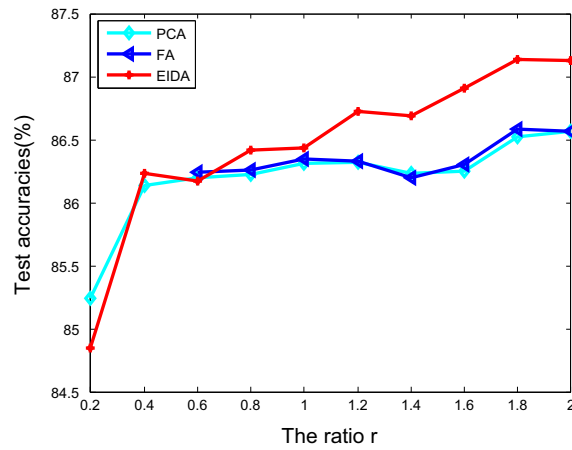
(a) Breast_cancer data with $m = 1$ (b) Twonorm data with $m = 2$

Fig. 6. The variation of the ratio of likelihood value and MI value in our method with the iteration in our experiment for the Breast_cancer and Twonorm datasets. Here m denotes the size of subspace.



(a) The results of all methods, i.e., PCA, FA, LDA, MMLDA, IDA, MIM and EIDA



(b) The results of PCA, FA and EIDA

Fig. 7. The variation of the classification accuracies with the ratios of numbers of training samples in each class to sample dimension on the synthetic dataset. Here the number of classes $K = 2$, the sample dimension $d = 50$, the number of total samples $n = 500$, the number of training samples in the k th class N_k decreasing from $n/K = 250$ with $k = 1, 2$, and the size of subspace $m = 1$.

$r \geq 2.75$ on Ringnorm data. When $r \leq 1.75$ on Ringnorm data, the accuracies of PCA, FA and MMLDA are better than that of EIDA. Fig. 8(c) and (d) show the variation of the classification accuracies with r on Waveform data, where Fig. 8(c) gives the results with $0.5 \leq r \leq 60$, Fig. 8(d) gives the results with $0.5 \leq r \leq 3$ to show the differences among all methods more clearly. In this experiment on Waveform data, $m = 1$, LDA, MMLDA, IDA and MIM cannot work when $r < 2$. It can be seen from Fig. 8(c) and (d), the proposed method, EIDA, outperforms other methods when $r \geq 0.75$ on Waveform data. When $r = 0.5$ on Waveform data, the accuracies of PCA and FA are a little better than that of EIDA.

In summary, the proposed method outperforms the other approaches in terms of high accuracies and low sensitivities to the training data sizes, especially on the condition of small training data. Therefore, the robustness of the proposed method is validated.

5.6. Computation time

In this subsection, we compare the dimensionality reduction methods from a computation time point of view. The experiments

of all algorithms are conducted using a PC with Intel(R) Core i5-4590 CPU of 3.3 GHz and memory of 8 GB. The program codes are written in MATLAB. Here we select Heart and Ringnorm datasets as two examples. Tables 12 and 13 show the computation time of the dimensionality reduction methods on these two datasets.

From Tables 12 and 13, we can see that the computation burden of EIDA is smaller than those of IDA and MIM, and larger than those of FA, PCA, LDA and MMLDA.

6. Conclusion

In this paper, we propose a novel feature transformation method for supervised dimensionality reduction. The proposed method describes the subspace statistical structure of the observed data via linear statistical model and further maximizes the between-class separability via information-theoretic criterion. As a result, the optimal transformation matrix and noise variance can be obtained via the joint optimization of MI function and log-likelihood function. To validate the proposed method, it is compared with FA, PCA, LDA,

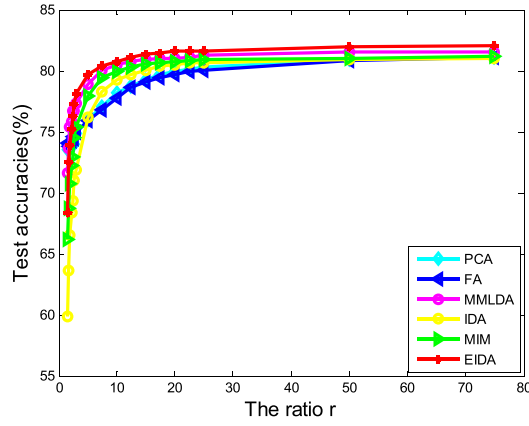
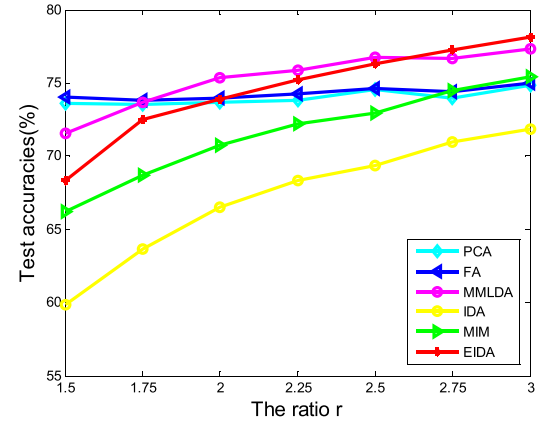
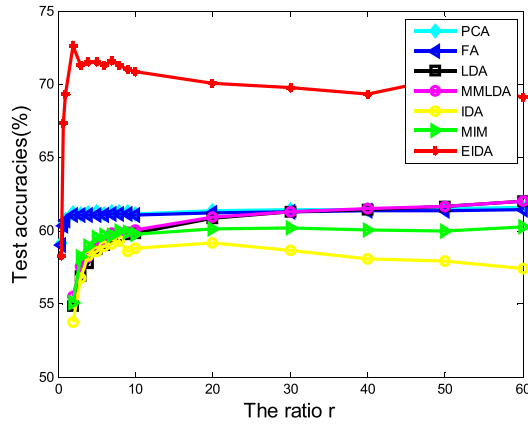
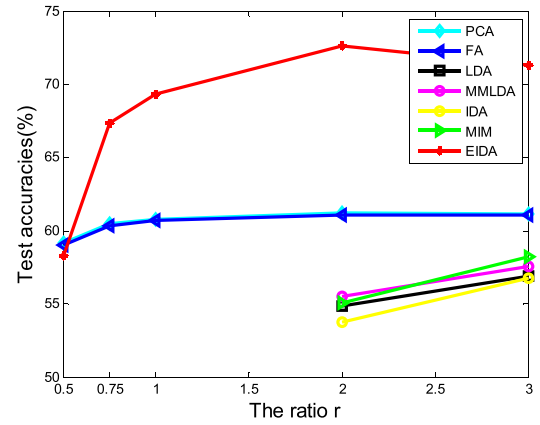
(a) The results on Ringnorm ($1.5 \leq r \leq 75$)(b) The results on Ringnorm ($1.5 \leq r \leq 3$)(c) The results on Waveform ($0.5 \leq r \leq 60$)(d) The results on Waveform ($0.5 \leq r \leq 3$)

Fig. 8. The variation of the classification accuracy with the ratio of the smallest number of training samples in each class to the sample dimension on Ringnorm and Waveform datasets. Here the size of subspace $m = 2$ for Ringnorm dataset and $m = 1$ for Waveform dataset.

Table 12
Computation time (s) of the dimensionality reduction methods on Heart data.

Method		Size of subspace (m)				
		1	2	3	4	5
FA	L	0.021499	0.021382	0.024661	0.023831	0.02462
	Q	0.021580	0.021445	0.024571	0.024032	0.023925
PCA	L	0.009734	0.008482	0.008977	0.008582	0.009159
	Q	0.008445	0.009761	0.009173	0.009095	0.008429
LDA	L	0.039611	–	–	–	–
	Q	0.056877	–	–	–	–
MMLDA	L	0.051901	0.052614	0.052184	0.053196	0.051552
	Q	0.069972	0.078842	0.077374	0.079552	0.079551
IDA	L	0.219417	0.382612	0.545834	0.740818	0.934114
	Q	0.231282	0.39981	0.573280	0.754107	0.956137
MIM	L	4.694605	6.106519	6.562560	6.654656	6.994241
	Q	4.914086	6.149458	6.762500	6.571009	7.050179
EIDA	L	0.074432	0.077081	0.076736	0.075383	0.074869
	Q	0.076663	0.076466	0.075259	0.075402	0.074740

Table 13
Computation time (s) of the dimensionality reduction methods on Ringnorm data.

Method		Size of subspace (m)				
		1	2	3	4	5
FA	L	0.025428	0.026261	0.027848	0.030150	0.030450
	Q	0.025022	0.025656	0.028157	0.029865	0.031085
PCA	L	0.010831	0.011545	0.011587	0.011582	0.012036
	Q	0.010743	0.011353	0.012047	0.011624	0.011346
LDA	L	0.042839	–	–	–	–
	Q	0.060060	–	–	–	–
MMLDA	L	0.043671	0.047021	0.047886	0.048059	0.048980
	Q	0.072270	0.093351	0.104545	0.121145	0.134307
IDA	L	0.226310	0.405013	0.589279	0.790472	1.003491
	Q	0.247785	0.431587	0.615861	0.804230	1.014992
MIM	L	5.879899	7.420181	7.551957	7.593585	8.164467
	Q	5.907079	7.123939	7.546449	7.621327	8.169589
EIDA	L	0.082294	0.081012	0.082175	0.084667	0.083574
	Q	0.081641	0.083670	0.085548	0.083832	0.085206

IDA, and MIM on synthetic dataset and benchmark datasets. Based on the experimental analysis on the classification accuracy, accuracy's standard deviation, and robustness to the size of training set, the proposed method outperforms the traditional feature transformation methods. Therefore, we infer that the proposed method should be considered as an alternative to the prevalent feature transformation methods.

Our method is the modified version of the traditional IDA method [7]. Both of IDA and our method assume that the data distribution in each class is a single Gaussian. In the future work, we can extend our method for the non-Gaussian case. In [33], a non-Gaussian FA model is proposed with the prior distribution on $\mathbf{y}$ being a GMM. We can use the similar prior in our linear statistical model. Since each class is a GMM with the above prior, we should also substitute the MIM objective [14] for the IDA objective [7] as the MI function. Nevertheless, in this way, the joint formulation for maximizing both MI function and log-likelihood function will be very complex.

Appendix

Proof of Theorem 1

Let $\text{tr}(\mathbf{D})$ represent the sum of the diagonal elements in the matrix $\mathbf{D}$. Since $\mathbf{F}_k = (\mathbf{A}\mathbf{A}^T + \Psi_k)^{-1}$, $\forall k = 1, 2, \dots, K$, we have $d\mathbf{F}_k = -\mathbf{F}_k d(\mathbf{A}\mathbf{A}^T + \Psi_k) \mathbf{F}_k = -\mathbf{F}_k d\Psi_k \mathbf{F}_k$. Thus we can derive that

$$\begin{aligned} dL_{\mathbf{A}, \Psi}(\mathbf{X}) &= -\frac{N_k}{2} \left[d(\ln |\mathbf{F}_k^{-1}|) + \text{tr} \left(d\mathbf{F}_k \left(\frac{1}{N_k} \sum_{i=1}^{N_k} \mathbf{x}_{ki} \mathbf{x}_{ki}^T \right) \right) \right] \\ &= -\frac{N_k}{2} \left[\text{tr} \left(\mathbf{F}_k d\Psi_k - \mathbf{F}_k d\Psi_k \mathbf{F}_k \left(\frac{1}{N_k} \sum_{i=1}^{N_k} \mathbf{x}_{ki} \mathbf{x}_{ki}^T \right) \right) \right] \end{aligned} \quad (23)$$

Consequently we have

$$\text{diag} \left(\frac{\partial L_{\mathbf{A}, \Psi}(\mathbf{X})}{\partial \Psi_k} \right) = \text{diag} \left(\frac{N_k}{2} \left[\mathbf{F}_k \left(\frac{1}{N_k} \sum_{i=1}^{N_k} \mathbf{x}_{ki} \mathbf{x}_{ki}^T \right) \mathbf{F}_k - \mathbf{F}_k \right] \right) \quad (24)$$

Proof of Theorem 2

According to the MI expression, we observe that

$$\begin{aligned} dI_{\mathbf{A}, \Psi}(\mathbf{C}, \mathbf{Y}) &= \frac{1}{2} \left[d(\log |\mathbf{A}^T \Omega \mathbf{A}|) - d \left(\sum_{k=1}^K w_k \log |\mathbf{A}^T \Omega_k \mathbf{A}| \right) \right] \\ &= \frac{1}{2} \left[\text{tr} \left(\mathbf{A} (\mathbf{A}^T \Omega \mathbf{A})^{-1} \mathbf{A}^T d\Omega \right) - \text{tr} \left(\mathbf{A} (\mathbf{A}^T \Omega_k \mathbf{A})^{-1} \mathbf{A}^T d\Omega_k \right) \right] \\ &= \frac{1}{2} \text{tr} \left(\mathbf{A} (\mathbf{A}^T \Omega \mathbf{A})^{-1} \mathbf{A}^T w_k d(\mathbf{F}_k \Sigma_{\mathbf{x}k} \mathbf{F}_k) \right) \\ &\quad + \frac{1}{2} \text{tr} \left(\mathbf{A} (\mathbf{A}^T \Omega \mathbf{A})^{-1} \mathbf{A}^T \sum_{k=1}^K w_k d \left[(\mu_k - \mu)(\mu_k - \mu)^T \right] \right) \\ &\quad - \frac{1}{2} \text{tr} \left(\mathbf{A} (\mathbf{A}^T \Omega_k \mathbf{A})^{-1} \mathbf{A}^T d(\mathbf{F}_k \Sigma_{\mathbf{x}k} \mathbf{F}_k) \right) \end{aligned} \quad (25)$$

Let $\mathbf{O} = \mathbf{A}(\mathbf{A}^T \Omega \mathbf{A})^{-1} \mathbf{A}^T$, the first term in Eq. (25) can be reformulated as

$$\begin{aligned} \frac{1}{2} \text{tr}(\mathbf{O} w_k d(\mathbf{F}_k \Sigma_{\mathbf{x}k} \mathbf{F}_k)) &= \frac{1}{2} w_k \text{tr}(\mathbf{O} (d\mathbf{F}_k \Sigma_{\mathbf{x}k} \mathbf{F}_k + \mathbf{F}_k \Sigma_{\mathbf{x}k} d\mathbf{F}_k)) \\ &= -\frac{1}{2} w_k \text{tr}(\mathbf{O} (\mathbf{F}_k d\Psi_k \mathbf{F}_k \Sigma_{\mathbf{x}k} \mathbf{F}_k \\ &\quad + \mathbf{F}_k \Sigma_{\mathbf{x}k} \mathbf{F}_k d\Psi_k \mathbf{F}_k)) \end{aligned} \quad (26)$$

Considering that $d(\mu_k - \mu) = d(\mathbf{F}_k \mu_{\mathbf{x}k} - \sum_{j=1}^K w_j \mathbf{F}_j \mu_{\mathbf{x}j}) = (w_k - 1) \mathbf{F}_k d\Psi_k \mathbf{F}_k \mu_{\mathbf{x}k}$, the second term in Eq. (25) can be reformulated as

$$\begin{aligned} &\frac{1}{2} \text{tr} \left(\mathbf{O} \sum_{k=1}^K w_k d \left[(\mu_k - \mu)(\mu_k - \mu)^T \right] \right) \\ &= \sum_{k=1}^K w_k \text{tr} \left((\mu_k - \mu)^T \mathbf{O} d(\mu_k - \mu) \right) \\ &= -w_k \text{tr} \left((\mu_k - \mu)^T \mathbf{O} \mathbf{F}_k d\Psi_k \mathbf{F}_k \mu_{\mathbf{x}k} \right) \end{aligned} \quad (27)$$

Let $\mathbf{O}_k = \mathbf{A}(\mathbf{A}^T \Omega_k \mathbf{A})^{-1} \mathbf{A}^T$, the third term in Eq. (25) can be reformulated as

$$\begin{aligned} &\frac{1}{2} \text{tr}(\mathbf{O}_k d(\mathbf{F}_k \Sigma_{\mathbf{x}k} \mathbf{F}_k)) \\ &= -\frac{1}{2} \text{tr}(\mathbf{O}_k (\mathbf{F}_k d\Psi_k \mathbf{F}_k \Sigma_{\mathbf{x}k} \mathbf{F}_k + \mathbf{F}_k \Sigma_{\mathbf{x}k} \mathbf{F}_k d\Psi_k \mathbf{F}_k)) \end{aligned} \quad (28)$$

Substituting Eqs. (26)–(28) into Eq. (25), and making use of $\Omega_k = \mathbf{F}_k \Sigma_{\mathbf{x}k} \mathbf{F}_k$, we finally have

$$\begin{aligned} &\text{diag} \left(\frac{\partial I_{\mathbf{A}, \Psi}(\mathbf{C}, \mathbf{Y})}{\partial \Psi_k} \right) \\ &= \text{diag} \left(-w_k \mathbf{F}_k \mathbf{O} \Omega_k - w_k \mathbf{F}_k \mathbf{O} (\mu_k - \mu) \mu_k^T + w_k \mathbf{F}_k \mathbf{O}_k \Omega_k \right) \\ &= \text{diag} \left(w_k \mathbf{F}_k \left[\mathbf{O} (\mu - \mu_k) \mu_k^T - (\mathbf{O} - \mathbf{O}_k) \Omega_k \right] \right) \end{aligned} \quad (29)$$

Proof of Theorem 3

According to the log-likelihood expression, we can derive that

$$\begin{aligned} dL_{\mathbf{A}, \Psi}(\mathbf{X}) &= -\frac{1}{2} \sum_{k=1}^K N_k \left[\text{tr} \left(\mathbf{F}_k (d\mathbf{A}\mathbf{A}^T + \mathbf{A}d\mathbf{A}^T) + d\mathbf{F}_k \left(\frac{1}{N_k} \sum_{i=1}^{N_k} \mathbf{x}_{ki} \mathbf{x}_{ki}^T \right) \right) \right] \\ &= -\frac{1}{2} \sum_{k=1}^K N_k \left[\text{tr} \left(\mathbf{F}_k (d\mathbf{A}\mathbf{A}^T + \mathbf{A}d\mathbf{A}^T) - \mathbf{F}_k (d\mathbf{A}\mathbf{A}^T + \mathbf{A}d\mathbf{A}^T) \mathbf{F}_k \left(\frac{1}{N_k} \sum_{i=1}^{N_k} \mathbf{x}_{ki} \mathbf{x}_{ki}^T \right) \right) \right] \end{aligned} \quad (30)$$

From which it follows that

$$\frac{\partial L_{\mathbf{A}, \Psi}(\mathbf{X})}{\partial \mathbf{A}} = \sum_{k=1}^K N_k \left[\mathbf{F}_k \left(\frac{1}{N_k} \sum_{i=1}^{N_k} \mathbf{x}_{ki} \mathbf{x}_{ki}^T \right) \mathbf{F}_k - \mathbf{F}_k \right] \mathbf{A} \quad (31)$$

Proof of Theorem 4

According to the MI expression, we can derive that

$$\begin{aligned} dI_{\mathbf{A}, \Psi}(\mathbf{C}, \mathbf{Y}) &= \frac{1}{2} \left\{ \text{tr} \left((\mathbf{A}^T \Omega \mathbf{A})^{-1} d(\mathbf{A}^T \Omega \mathbf{A}) \right) \right. \\ &\quad \left. - \sum_{k=1}^K w_k \text{tr} \left((\mathbf{A}^T \Omega_k \mathbf{A})^{-1} d(\mathbf{A}^T \Omega_k \mathbf{A}) \right) \right\} \\ &= \frac{1}{2} \sum_{k=1}^K w_k \text{tr} \left((\mathbf{A}^T \Omega \mathbf{A})^{-1} d(\mathbf{A}^T \Omega_k \mathbf{A}) \right) \\ &\quad + \frac{1}{2} \sum_{k=1}^K w_k \text{tr} \left((\mathbf{A}^T \Omega \mathbf{A})^{-1} d \left[(\mu_k - \mu)(\mu_k - \mu)^T \mathbf{A} \right] \right) \\ &\quad - \frac{1}{2} \sum_{k=1}^K w_k \text{tr} \left((\mathbf{A}^T \Omega_k \mathbf{A})^{-1} d(\mathbf{A}^T \Omega_k \mathbf{A}) \right) \end{aligned} \quad (32)$$

Let $\mathbf{G}_k = \mathbf{A}^T \mathbf{F}_k$, we obtain $d\mathbf{G}_k = (d\mathbf{A})^T \mathbf{F}_k - \mathbf{A}^T \mathbf{F}_k d\mathbf{A} \mathbf{A}^T \mathbf{F}_k - \mathbf{A}^T \mathbf{F}_k \mathbf{A} (d\mathbf{A})^T \mathbf{F}_k$. Then the first term in Eq. (32) can be expressed as

$$\begin{aligned}
& \frac{1}{2} \sum_{k=1}^K w_k \text{tr} \left((\mathbf{A}^T \mathbf{\Omega} \mathbf{A})^{-1} d (\mathbf{A}^T \mathbf{\Omega}_k \mathbf{A}) \right) \\
&= \frac{1}{2} \sum_{k=1}^K w_k \text{tr} \left((\mathbf{A}^T \mathbf{\Omega} \mathbf{A})^{-1} d (\mathbf{G}_k \mathbf{\Sigma}_{\mathbf{x}k} \mathbf{G}_k^T) \right) \\
&= \frac{1}{2} \sum_{k=1}^K w_k \text{tr} \left((\mathbf{A}^T \mathbf{\Omega} \mathbf{A})^{-1} \left[d \mathbf{G}_k \mathbf{\Sigma}_{\mathbf{x}k} \mathbf{G}_k^T + \mathbf{G}_k \mathbf{\Sigma}_{\mathbf{x}k} (d \mathbf{G}_k)^T \right] \right) \\
&= \sum_{k=1}^K w_k \text{tr} \left(\mathbf{\Omega}_k \mathbf{A} (\mathbf{A}^T \mathbf{\Omega} \mathbf{A})^{-1} (d \mathbf{A})^T - \mathbf{A}^T \mathbf{\Omega}_k \mathbf{O} \mathbf{F}_k d \mathbf{A} - \mathbf{\Omega}_k \mathbf{O} \mathbf{F}_k \mathbf{A} (d \mathbf{A})^T \right) \quad (33)
\end{aligned}$$

And the second term in Eq. (32) can be expressed as

$$\begin{aligned}
& \frac{1}{2} \sum_{k=1}^K w_k \text{tr} \left((\mathbf{A}^T \mathbf{\Omega} \mathbf{A})^{-1} d \left[\mathbf{A}^T (\mu_k - \mu) (\mu_k - \mu)^T \mathbf{A} \right] \right) \\
&= \sum_{k=1}^K w_k \text{tr} \left((\mu_k - \mu)^T \mathbf{A} (\mathbf{A}^T \mathbf{\Omega} \mathbf{A})^{-1} d \left[\mathbf{A}^T (\mu_k - \mu) \right] \right) \\
&= \sum_{k=1}^K w_k \text{tr} \left((\mu_k - \mu)^T \mathbf{A} (\mathbf{A}^T \mathbf{\Omega} \mathbf{A})^{-1} \left(d \mathbf{G}_k \mu_{\mathbf{x}k} - \sum_{j=1}^K w_j d \mathbf{G}_j \mu_{\mathbf{x}j} \right) \right) \\
&= \sum_{k=1}^K w_k \text{tr} \left(\begin{array}{l} \mu_k (\mu_k - \mu)^T \mathbf{A} (\mathbf{A}^T \mathbf{\Omega} \mathbf{A})^{-1} (d \mathbf{A})^T - \mathbf{A}^T \mu_k (\mu_k - \mu)^T \mathbf{O} \mathbf{F}_k d \mathbf{A} \\ - \mu_k (\mu_k - \mu)^T \mathbf{O} \mathbf{F}_k \mathbf{A} (d \mathbf{A})^T - \mu (\mu_k - \mu)^T \mathbf{A} (\mathbf{A}^T \mathbf{\Omega} \mathbf{A})^{-1} (d \mathbf{A})^T \end{array} \right) \quad (34)
\end{aligned}$$

Since the third term has the same structure with the first term, the third term in Eq. (32) can be expressed as

$$\begin{aligned}
& \frac{1}{2} \sum_{k=1}^K w_k \text{tr} \left((\mathbf{A}^T \mathbf{\Omega}_k \mathbf{A})^{-1} d (\mathbf{A}^T \mathbf{\Omega}_k \mathbf{A}) \right) \\
&= \sum_{k=1}^K w_k \text{tr} \left(\mathbf{\Sigma}_{\mathbf{x}k} \mathbf{G}_k^T (\mathbf{A}^T \mathbf{\Omega}_k \mathbf{A})^{-1} d \mathbf{G}_k \right) \\
&= \sum_{k=1}^K w_k \text{tr} \left(\mathbf{\Omega}_k \mathbf{A} (\mathbf{A}^T \mathbf{\Omega}_k \mathbf{A})^{-1} (d \mathbf{A})^T - \mathbf{A}^T \mathbf{\Omega}_k \mathbf{O}_k \mathbf{F}_k d \mathbf{A} - \mathbf{\Omega}_k \mathbf{O}_k \mathbf{F}_k \mathbf{A} (d \mathbf{A})^T \right) \quad (35)
\end{aligned}$$

Let $\mathbf{P}_k = \mathbf{F}_k \mathbf{O} (\mu - \mu_k) \mu_k^T$, and $\mathbf{M}_k = \mathbf{F}_k (\mathbf{O} - \mathbf{O}_k) \mathbf{\Omega}_k$. Substituting Eqs. (33)–(35) into Eq. (32), we can get

$$\begin{aligned}
\frac{\partial I_{\mathbf{A}, \Psi}(\mathbf{C}, \mathbf{Y})}{\partial \mathbf{A}} &= \sum_{k=1}^K w_k \left[\mathbf{\Omega}_k \mathbf{A} (\mathbf{A}^T \mathbf{\Omega} \mathbf{A})^{-1} - \mathbf{F}_k \mathbf{O} \mathbf{\Omega}_k \mathbf{A} - \mathbf{\Omega}_k \mathbf{O} \mathbf{F}_k \mathbf{A} + \mu_k (\mu_k - \mu)^T \mathbf{A} (\mathbf{A}^T \mathbf{\Omega} \mathbf{A})^{-1} \right. \\
&\quad - \mathbf{F}_k \mathbf{O} (\mu_k - \mu) \mu_k^T \mathbf{A} - \mu_k (\mu_k - \mu)^T \mathbf{O} \mathbf{F}_k \mathbf{A} - \mu (\mu_k - \mu)^T \mathbf{A} (\mathbf{A}^T \mathbf{\Omega} \mathbf{A})^{-1} \\
&\quad \left. - \mathbf{\Omega}_k \mathbf{A} (\mathbf{A}^T \mathbf{\Omega}_k \mathbf{A})^{-1} + \mathbf{F}_k \mathbf{O}_k \mathbf{\Omega}_k \mathbf{A} + \mathbf{\Omega}_k \mathbf{O}_k \mathbf{F}_k \mathbf{A} \right] \\
&= \mathbf{\Omega} \mathbf{A} (\mathbf{A}^T \mathbf{\Omega} \mathbf{A})^{-1} - \sum_{k=1}^K w_k \mathbf{\Omega}_k \mathbf{A} (\mathbf{A}^T \mathbf{\Omega}_k \mathbf{A})^{-1} \\
&\quad + \sum_{k=1}^K w_k (\mathbf{P}_k + \mathbf{P}_k^T - \mathbf{M}_k - \mathbf{M}_k^T) \mathbf{A} \quad (36)
\end{aligned}$$

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